In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.read_csv(r"C:\Users\tjyot\Downloads\archive1\IMDb Movies India.csv", encoding='ISO-8859-1')
```

In [3]:
```
df
```

Out[3]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
0		NaN	NaN	Drama	NaN	NaN	J.S. Randhawa	Manmauji	Birbal	Rajendra Bhatia
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
2	#Homecoming	(2021)	90 min	Drama, Musical	NaN	NaN	Soumyajit Majumdar	Sayani Gupta	Plabita Borthakur	Roy Angana
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
4	...And Once Again	(2010)	105 min	Drama	NaN	NaN	Amol Palekar	Rajat Kapoor	Rituparna Sengupta	Antara Mali
...	...	...	...	...	...	...	...	...	...	...
15504	Zulm Ko Jala Doonga	(1988)	NaN	Action	4.6	11	Mahendra Shah	Naseeruddin Shah	Sumeet Saigal	Suparna Anand
15505	Zulmi	(1999)	129 min	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani
15506	Zulmi Raj	(2005)	NaN	Action	NaN	NaN	Kiran Thej	Sangeeta Tiwari	NaN	NaN
15507	Zulmi Shikari	(1988)	NaN	Action	NaN	NaN	NaN	NaN	NaN	NaN
15508	Zulm-O-Sitam	(1998)	130 min	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja

15509 rows × 10 columns

In [4]:
```
df.shape
```

Out[4]:
```
(15509, 10)
```

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15509 entries, 0 to 15508
Data columns (total 10 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   Name      15509 non-null  object 
 1   Year      14981 non-null  object 
 2   Duration  7240 non-null   object 
 3   Genre     13632 non-null  object 
 4   Rating    7919 non-null   float64
 5   Votes     7920 non-null   object 
 6   Director  14984 non-null  object 
 7   Actor 1   13892 non-null  object 
 8   Actor 2   13125 non-null  object 
 9   Actor 3   12365 non-null  object 
dtypes: float64(1), object(9)
memory usage: 1.2+ MB
```

In [6]:
```
df.isna().sum()
```

Out[6]:
```
Name          0
Year        528
Duration   8269
Genre      1877
Rating     7590
Votes      7589
Director    525
Actor 1    1617
Actor 2    2384
Actor 3    3144
dtype: int64
```

In [7]:
```
df.duplicated().sum()
```

Out[7]:
```
6
```

In [8]:
```
df.dropna(inplace=True)
```

In [9]:
```
df.drop_duplicates(inplace=True)
```

In [10]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5659 entries, 1 to 15508
Data columns (total 10 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   Name      5659 non-null   object 
 1   Year      5659 non-null   object 
 2   Duration  5659 non-null   object 
 3   Genre     5659 non-null   object 
 4   Rating    5659 non-null   float64
 5   Votes     5659 non-null   object 
 6   Director  5659 non-null   object 
 7   Actor 1   5659 non-null   object 
 8   Actor 2   5659 non-null   object 
 9   Actor 3   5659 non-null   object 
dtypes: float64(1), object(9)
memory usage: 486.3+ KB
```

In [11]:
```
df.shape
```

Out[11]:
```
(5659, 10)
```

In [12]:
```
df
```

Out[12]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	(2019)	109 min	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	(2019)	110 min	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	(1997)	147 min	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	(2005)	142 min	Drama, Romance, War	7.4	1,086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	(2012)	82 min	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15493	Zubaan	(2015)	115 min	Drama	6.1	408	Mozez Singh	Vicky Kaushal	Sarah Jane Dias	Raaghavv Chanana
15494	Zubeidaa	(2001)	153 min	Biography, Drama, History	6.2	1,496	Shyam Benegal	Karisma Kapoor	Rekha	Manoj Bajpayee
15503	Zulm Ki Zanjeer	(1989)	125 min	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15505	Zulmi	(1999)	129 min	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani
15508	Zulm-O-Sitam	(1998)	130 min	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja

5659 rows × 10 columns

In [13]:
```
import pandas as pd

# Ensure the column is treated as a string before applying .str methods
df['Year'] = df['Year'].astype(str).str.replace(r'\(|\)', '', regex=True).astype(int)
df['Duration'] = pd.to_numeric(df['Duration'].astype(str).str.replace(' min', ''))
df['Votes'] = pd.to_numeric(df['Votes'].astype(str).str.replace(',', ''))
```

In [14]:
```
df.dtypes
```

Out[14]:
```
Name         object
Year          int32
Duration      int64
Genre        object
Rating      float64
Votes         int64
Director     object
Actor 1      object
Actor 2      object
Actor 3      object
dtype: object
```

In [15]:
```
df
```

Out[15]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	2019	109	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	2019	110	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	1997	147	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	2005	142	Drama, Romance, War	7.4	1086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	2012	82	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15493	Zubaan	2015	115	Drama	6.1	408	Mozez Singh	Vicky Kaushal	Sarah Jane Dias	Raaghavv Chanana
15494	Zubeidaa	2001	153	Biography, Drama, History	6.2	1496	Shyam Benegal	Karisma Kapoor	Rekha	Manoj Bajpayee
15503	Zulm Ki Zanjeer	1989	125	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15505	Zulmi	1999	129	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani
15508	Zulm-O-Sitam	1998	130	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja

5659 rows × 10 columns

In [16]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5659 entries, 1 to 15508
Data columns (total 10 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      5659 non-null   object
 1   Year      5659 non-null   int32
 2   Duration  5659 non-null   int64
 3   Genre     5659 non-null   object
 4   Rating    5659 non-null   float64
 5   Votes     5659 non-null   int64
 6   Director  5659 non-null   object
 7   Actor 1   5659 non-null   object
 8   Actor 2   5659 non-null   object
 9   Actor 3   5659 non-null   object
dtypes: float64(1), int32(1), int64(2), object(6)
memory usage: 464.2+ KB
```

In [17]:
```
df
```

Out[17]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
1	#Gadhvi (He thought he was Gandhi)	2019	109	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid
3	#Yaaram	2019	110	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor
5	...Aur Pyaar Ho Gaya	1997	147	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor
6	...Yahaan	2005	142	Drama, Romance, War	7.4	1086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma
8	?: A Question Mark	2012	82	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia
...	...	...	...	...	...	...	...	...	...	...
15493	Zubaan	2015	115	Drama	6.1	408	Mozez Singh	Vicky Kaushal	Sarah Jane Dias	Raaghavv Chanana
15494	Zubeidaa	2001	153	Biography, Drama, History	6.2	1496	Shyam Benegal	Karisma Kapoor	Rekha	Manoj Bajpayee
15503	Zulm Ki Zanjeer	1989	125	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth
15505	Zulmi	1999	129	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani
15508	Zulm-O-Sitam	1998	130	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja

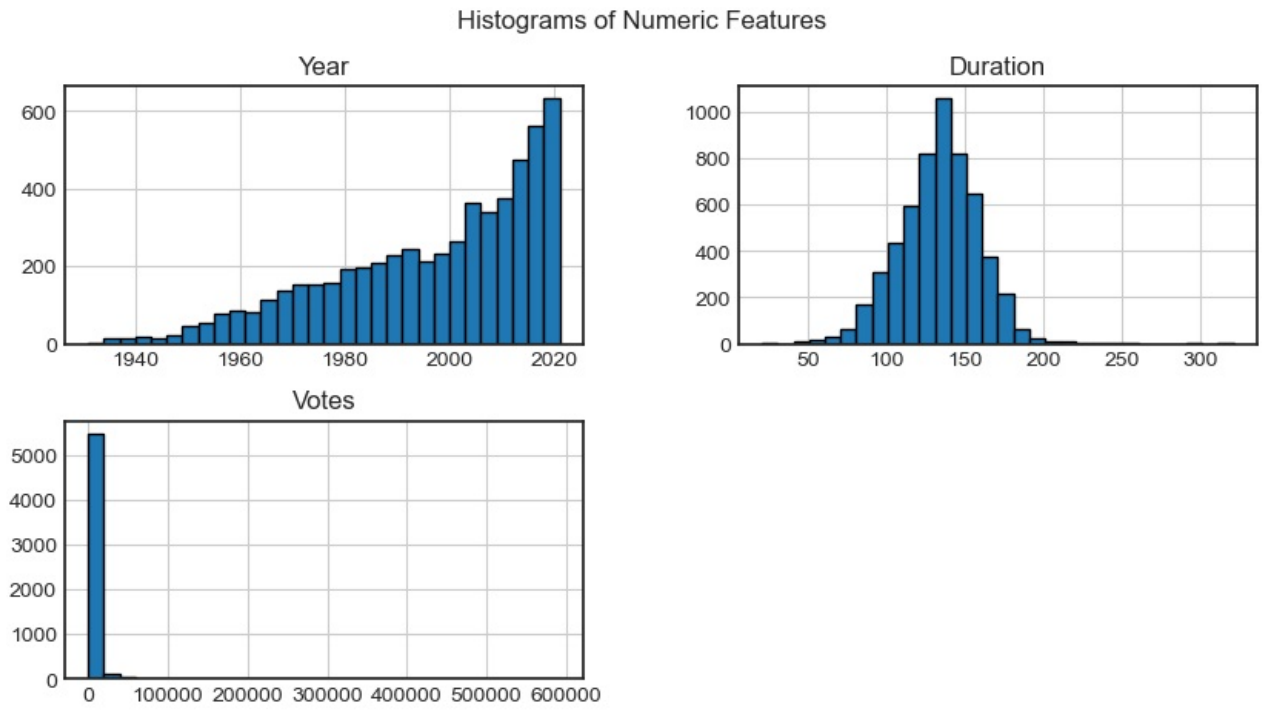
5659 rows × 10 columns

In [18]:
```
df.describe(include='all')
```

Out[18]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
count	5659	5659.000000	5659.000000	5659	5659.000000	5659.000000	5659	5659	5659	5659
unique	5385	NaN	NaN	376	NaN	NaN	2431	1960	2321	2556
top	Baazi	NaN	NaN	Drama	NaN	NaN	David Dhawan	Amitabh Bachchan	Rekha	Pran
freq	4	NaN	NaN	844	NaN	NaN	41	104	52	57
mean	NaN	1996.247570	133.439124	NaN	5.898533	2697.649585	NaN	NaN	NaN	NaN
std	NaN	19.741839	25.319939	NaN	1.381165	13651.503584	NaN	NaN	NaN	NaN
min	NaN	1931.000000	21.000000	NaN	1.100000	5.000000	NaN	NaN	NaN	NaN
25%	NaN	1983.000000	119.000000	NaN	5.000000	30.000000	NaN	NaN	NaN	NaN
50%	NaN	2002.000000	135.000000	NaN	6.100000	131.000000	NaN	NaN	NaN	NaN
75%	NaN	2013.000000	150.000000	NaN	6.900000	922.500000	NaN	NaN	NaN	NaN
max	NaN	2021.000000	321.000000	NaN	10.000000	591417.000000	NaN	NaN	NaN	NaN

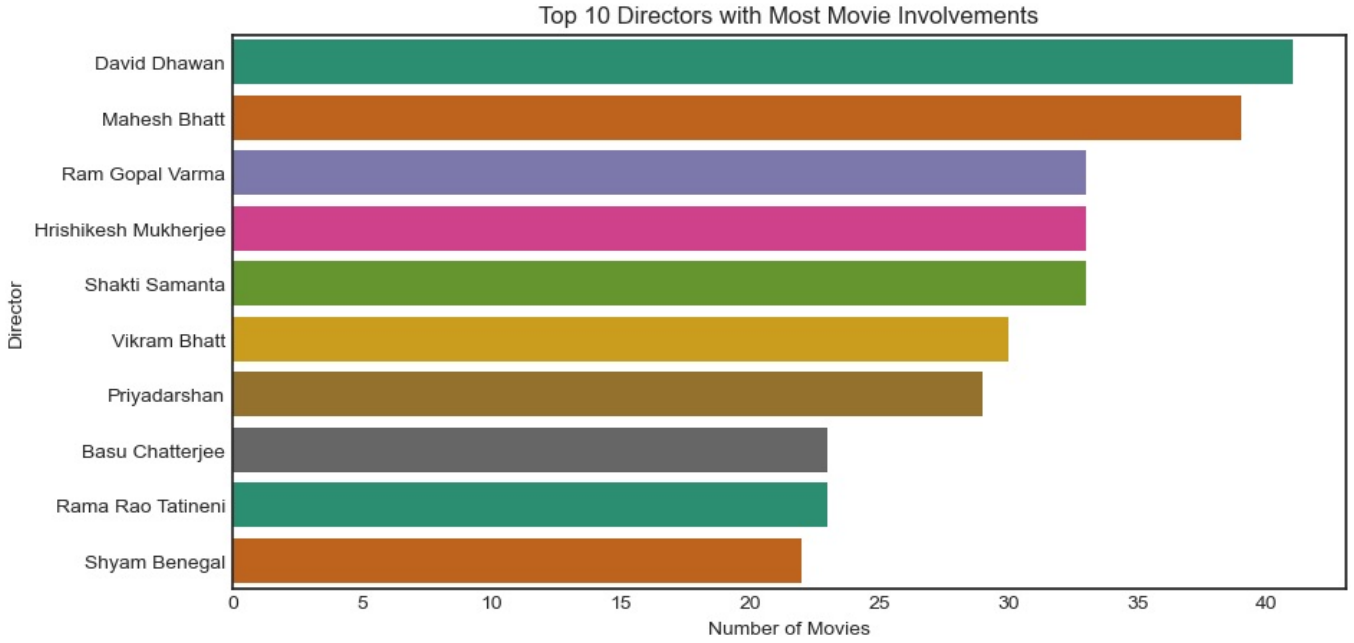
In [19]:
```
plt.style.use('seaborn-v0_8-white')
df[['Year', 'Duration', 'Votes']].hist(bins=30, edgecolor='black',figsize=(10, 5))
plt.suptitle('Histograms of Numeric Features')
plt.show()
```



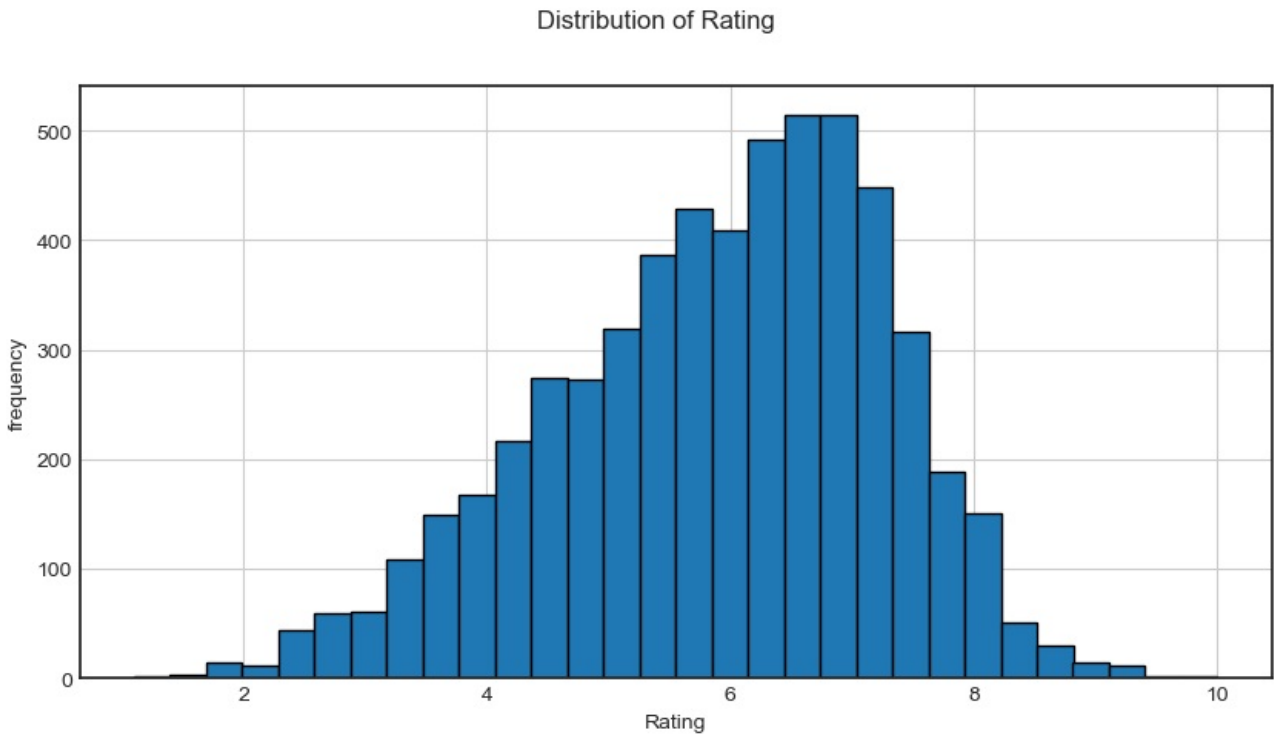
In [20]:
```
import matplotlib.pyplot as plt
import seaborn as sns

top_10_directors = df['Director'].value_counts(ascending=False).head(10)

plt.figure(figsize=(10, 5))
sns.barplot(x=top_10_directors.values, y=top_10_directors.index, hue=top_10_directors.index, palette='Dark2', l
plt.title('Top 10 Directors with Most Movie Involvements')
plt.xlabel('Number of Movies')
plt.ylabel('Director')
plt.show()
```



In [21]:
```
df['Rating'].hist(bins=30, edgecolor='black',figsize=(10, 5))
plt.suptitle('Distribution of Rating')
plt.xlabel('Rating')
plt.ylabel('frequency')
plt.show()
```



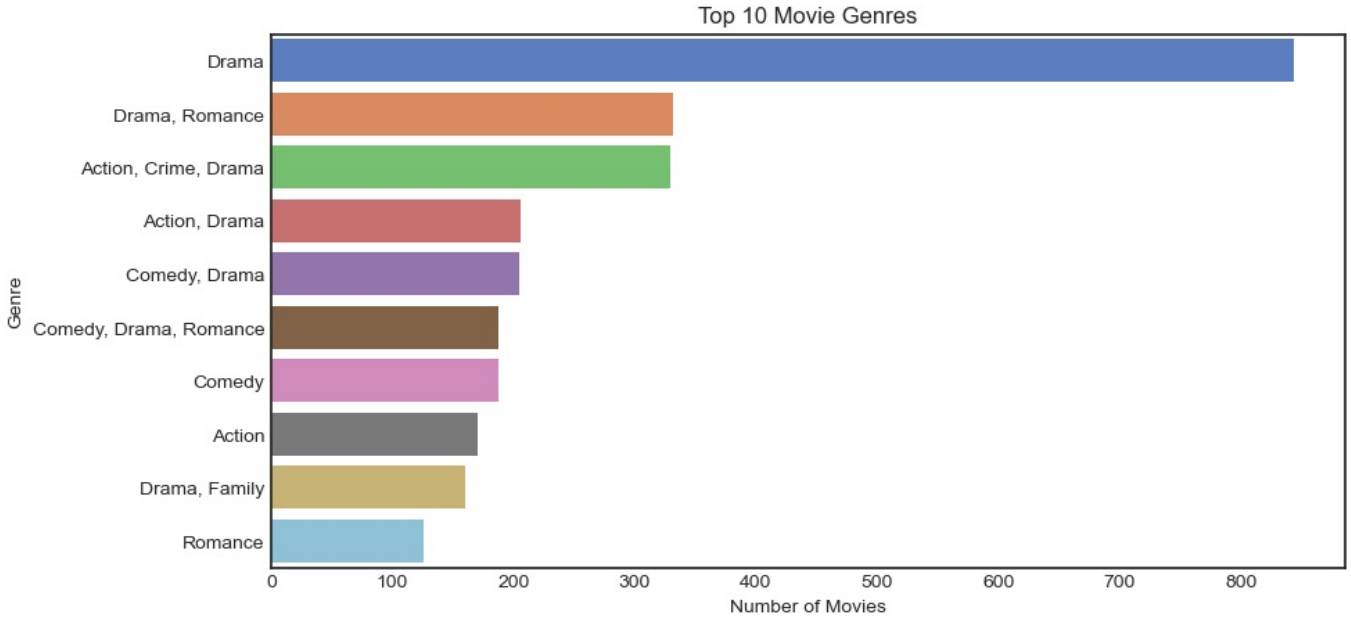
In [22]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Get top 10 genres
top_10_genres = df['Genre'].value_counts(ascending=False).head(10)

plt.figure(figsize=(10, 5))

# Fix: Assign `hue=top_10_genres.index` and set `legend=False`
sns.barplot(x=top_10_genres.values, y=top_10_genres.index, hue=top_10_genres.index, palette='muted', legend=Fals

plt.title('Top 10 Movie Genres')
plt.xlabel('Number of Movies')
plt.ylabel('Genre')
plt.show()
```



In [23]:
```
combined_actors = pd.concat([df['Actor 1'], df['Actor 2'], df['Actor 3']])
top_10_actors = combined_actors.value_counts().head(10)
top_10_actors
```

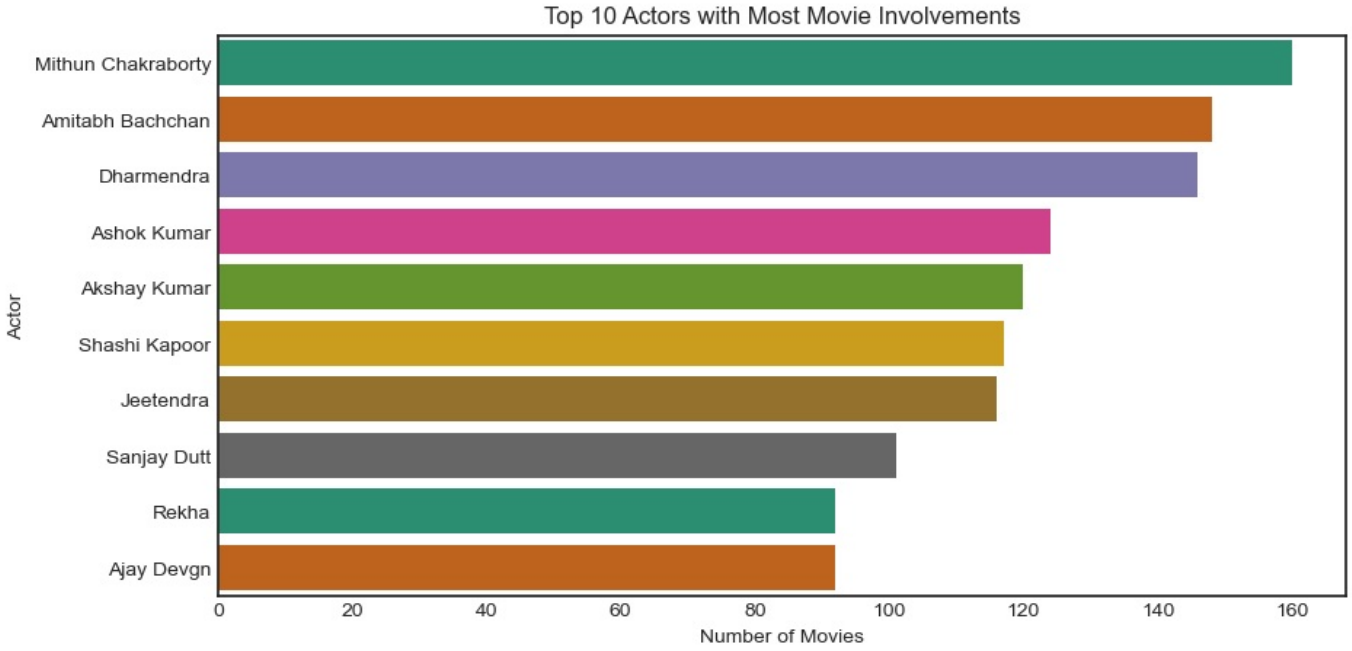
Out[23]:
```
Mithun Chakraborty    160
Amitabh Bachchan      148
Dharmendra            146
Ashok Kumar           124
Akshay Kumar          120
Shashi Kapoor         117
Jeetendra             116
Sanjay Dutt           101
Rekha                  92
Ajay Devgn             92
Name: count, dtype: int64
```

In [24]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 5))

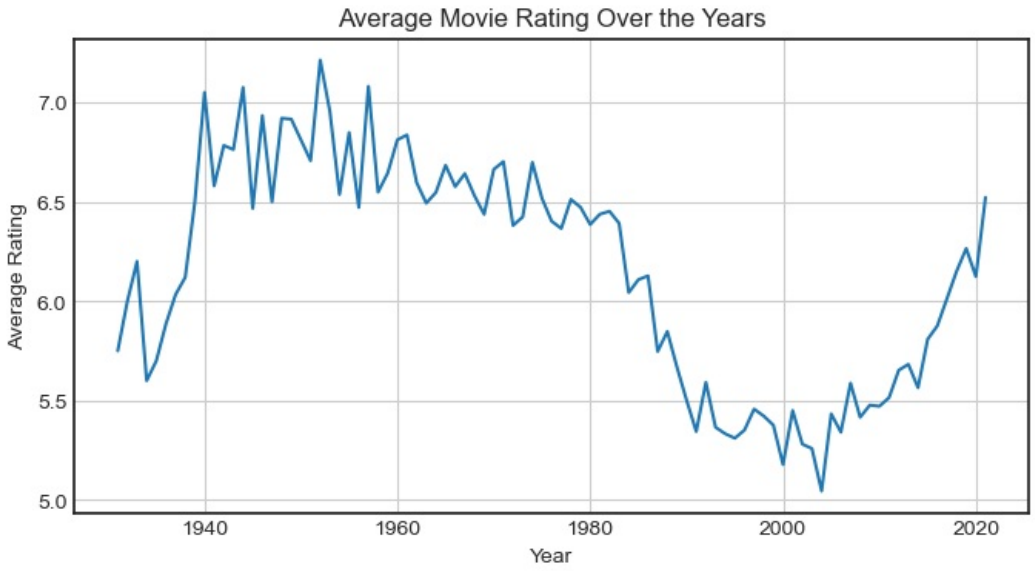
# Fix: Assign `hue=top_10_actors.index` and set `legend=False`
sns.barplot(x=top_10_actors.values, y=top_10_actors.index, hue=top_10_actors.index, palette='Dark2', legend=Fals

plt.title('Top 10 Actors with Most Movie Involvements')
plt.xlabel('Number of Movies')
plt.ylabel('Actor')
plt.show()
```



In [25]:
```
average_rating_per_year = df.groupby('Year')['Rating'].mean().reset_index()
average_rating_per_year.columns = ['Year', 'Average Rating']

plt.figure(figsize=(8, 4))
plt.plot(average_rating_per_year['Year'], average_rating_per_year['Average Rating'], linestyle='-')
plt.xlabel('Year')
plt.ylabel('Average Rating')
plt.title('Average Movie Rating Over the Years')
plt.grid(True)
plt.show()
```



In [26]:
```
high_rating_movies = df[df['Rating'] > 8.5]
high_rating_movies
```

Out[26]:

	Name	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3
967	An Insignificant Man	2016	96	Documentary, Thriller	8.7	1928	Khushboo Ranka	Vinay Shukla	Arvind Kejriwal	Saurabh Bhardwaj
1314	Ashok Vatika	2018	97	Drama	9.3	7	Rahul Mallick	Kunj Anand	Sanjay Bishnoi	Paras Zutshi
1341	Atal Faisla	2018	127	Drama	8.8	5	Abdul Sattar	Sahil Akhtar	Himayat Ali	Aman Jain
1432	Awasaan	2020	101	Drama	8.7	21	Rajat Kumar Chaturvedi	Arjun Chakrabarty	Trishala Idnani	Sarvagya Jain
1469	B for Bundelkhand	2017	117	Drama	8.8	5	Vishal Mourya	Bharat Chawla	Nemi Chandra Jha	Moumita Nandi
1709	Bahattar Hoorain	2019	120	Comedy	8.8	23	Sanjay Puran Singh Chauhan	Narottam Bain	Aamir Bashir	Pawan Malhotra
1729	Baikunth	2021	72	Family	9.3	29	Vishwa Bhanu	Vishwa Bhanu	Sangam Shukla	Vijay Thakur
2093	Bhagwan Parshuram	1970	136	Drama	8.7	10	Babubhai Mistry	Niranjan Sharma	Shalini	Trilok Kapoor
2437	Black Rose	2020	120	Thriller	8.6	38	Hemant N Mishra	Samiksha Bhatnagar	Rahul Chawla	Abhilash Dada
2709	Chaar Sahibzaade 2: Rise of Banda Singh Bahadur	2016	134	Animation	8.6	656	Harry Baweja	Harman Baweja	Pawan Kalra	Om Puri
2762	Chale Chalo: The Lunacy of Film Making	2003	150	Documentary	8.6	336	Satyajit Bhatkal	Ashutosh Gowariker	Aamir Khan	Gracy Singh
3224	Consequence Karma	2021	92	Drama, Thriller	9.0	223	Shadab Ahmad	Sidharth Bhardwaj	Monis Khan	Pooja Gupta
3802	Dhuusar	2019	106	Biography, Crime, Drama	8.9	47	Snehashish Mondal	Soumi Saha	Arpita Banerjee	Ritwik Bhowmik
4953	Gauru: Journey of Courage	2016	124	Family	8.8	85	Ramkishan Choyal	Jahangir Khan	Vedshree Rao	Ila Arun
5077	Gho Gho Rani	2019	105	History, Romance	9.4	47	Munni Pankaj	Nishi Neha Mishra	Pankaj Kamal	Akash Kumar
5125	God of gods	2019	90	Documentary	9.3	46	Venkatesh Bk	Tejaswini Manogna	Triyug Mantri	Raj Singh Verma
5410	Half Songs	2021	79	Music, Romance	9.7	7	Sriram Raja	Raj Banerjee	Emon Chatterjee	Purshottam Mulani
5980	I Am Offended	2015	102	Documentary	8.6	17	Jaideep Varma	Tanmay Bhat	Cyrus Broacha	Vir Das
6025	I'm in Love	2007	140	Action	9.2	36	Tom Alter	Bobby Darling	Anil Dhawan	Kiran Juneja
6743	Jihad	2017	144	Drama	8.6	47	Rakesh Parmaar	Farhad Ahmed Dehlvi	Alfeeya Donna	Ahmer Haider
6811	Josef - Born in Grace	2019	100	Drama	9.0	12	Susant Misra	Victor Banerjee	Subrat Dutta	Sudarshan Juyal

6852	June	2021	93	Drama	9.4	18	Suhrud Godbole	Vaibhav Khisti	Nilesh Divekar	Jitendra Joshi
6952	Kaadan	2021	170	Action, Drama, Fantasy	9.1	2219	Prabu Solomon	Raghu Babu	Rana Daggubati	Abudhar Al Hassan
6961	Kaagaz Ki Kashti	2017	127	Documentary	8.9	203	Brahmanand S. Singh	Ghulam Ali	Talat Aziz	Mahesh Bhatt
7824	Koshish	1972	125	Drama	8.6	1523	Gulzar	Sanjeev Kumar	Jaya Bachchan	Asrani
7939	Kutte Ki Dum	2018	143	Comedy	8.9	10	Sunil Patel	Raz Rehman Ali	Jass Boparai	Tanya Daang
8157	Last Chance: The Moment I Finally Embraced the...	2019	74	Comedy, Drama	8.8	10	Ronit Jadhav	Ronit Jadhav	Rohit D'Souza	Darshil Gandhi
8188	Leaving Home: The Life and Music of Indian Ocean	2008	115	Documentary, Music	8.9	94	Jaideep Varma	Kilam Amit	Asheem Chakravarty	Rahul Ram
8238	Little Baby	2019	105	Family	8.6	1417	Shekhar S. Jha	Priyanshu Chatterjee	Gulnaz Siganporia	Aadi
8339	Love Qubool Hai	2020	94	Drama, Romance	10.0	5	Saif Ali Sayeed	Ahaan Jha	Mahesh Narayan	Rajasree Rajakumari
8344	Love Sorries	2021	101	Comedy, Drama, Romance	9.3	79	Gautam Joshi	Prashant Chaubey	Puneet Chouksey	Amitabh Gupta
8466	Maassab (The Teacher)	2018	116	Drama	8.6	155	Aditya Om	Shiva Suryavanshi	Sheetal Singh	Manveer Choudhary
8474	Machaan	2021	124	Drama, Romance	8.6	93	Nitesh Tiwari	Pawan Waval	Richa Kalra	Prateek Goyal
9189	Mera Ramrajya Nai Kranti	2020	162	Drama	8.8	16	Shyam Bharteey	Shyam Bharteey	Ashwin Krishna Gaur	Aman Gupta
9504	Moral Nights	2021	93	Drama	9.0	37	S. Sunil	Kannanunni	Santhosh Keezhattoor	Sandra Larwin
9590	Mudda 370 J&K	2019	132	Action, Drama	8.6	699	Rakesh Sawant	Hiten Tejwani	Anjali Pandey	Manoj Joshi
9968	Naqab	1955	130	Romance	8.6	16	Lekhraj Bhakri	Shammi Kapoor	Madhubala	Ajit
10210	Nirmal Anand Ki Puppy	2019	100	Drama	9.1	7	Sandeep Mohan	Karanveer	Gillian Pinto	Khushboo Upadhyay
10217	Nirvana13	2011	95	Drama	9.1	99	Jitin Rawat	P.P. Bajaj	Anu Choudhury	Sandeep Kulkarni
10506	Pancham Unmixed: Mujhe Chalte Jaana Hai ...	2008	113	Documentary	8.6	43	Brahmanand S. Singh	Anju Bawra	Gulshan Bawra	Anand Bhosle
10862	Pinti Ka Sabun	2017	85	Drama	8.7	6	Pramod Pathak	Diamond Akhtar	Sukriti Chamoli	Satish Dhaulakhandi
11841	Rediscovering India	2017	124	Documentary	9.0	62	Meenal Dixit	Benny John	Benny John	Meenal Dixit
11843	Refl3ct	2021	65	Sci-Fi	9.3	467	Nikhil Mahar	Vijay Mahar	Vijay Mahar	Nikhil Mahar
11932	RK/RKAY	2021	95	Drama	8.7	12	Rajat Kapoor	Rajat Kapoor	Mallika Sherawat	Ranvir Shorey
12165	Sachin - A Billion Dreams	2017	138	Documentary, Sport	8.6	7711	James Erskine	Sachin Tendulkar	Mikhail Gandhi	Anjali Tendulkar
12673	Secrets of Sinauli	2021	56	Documentary, History	9.3	1373	Raghav Jairath	Manoj Bajpayee	R.S. Bhist	K.N. Dixit
13231	Sindhustan	2019	64	Documentary, Family, History	9.3	36	Sapna Bhavnani	Leila Advani	Laj Badlani	Chaho Bhara
13384	Soorarai Pottru	2020	153	Drama	9.1	83029	Sudha Kongara	Suriya	Madhavan	Paresh Rawal
13584	Sunset Bollywood	2005	60	Documentary	8.6	663	Komal Tolani	Amitabh Bachchan	Bhagyashree	Mahesh Bhatt
13676	T for Taj Mahal	2018	104	Drama	8.7	93	Kireet Khurana	Subrat Dutta	Pitobash	Manoj Pahwa
13766	Tank Cleaner	2021	155	Thriller	8.7	20	Parvinder Singh Wraich	Harkiran Kaur Wraich	Muntazir Ahmad	Sunny Gill Ambersariya
14071	The Flip Side: A Truth That Could Not Reach You	2015	48	Documentary	9.2	97	Utpal Kalal	Subramanian Swamy	Shiva Shambh Bhatt	Asaram Bapu

14198	The Pickup Artist	2019	111	Crime, Mystery, Thriller	8.7	85	Rohit Arora	Saasha Aery	Sidharth Bhardwaj	Anahita Bhooshan
14209	The Pushkar Lodge	2020	115	Drama	8.6	14	Vijay Suthar	Anil Chahar	Sachin Chaudhary	Yashvi Chhabra
14222	The Reluctant Crime	2020	113	Drama	9.4	16	Arvind Pratap	Dharmendra Ahir	Awanish Kotnal	Rakhi Mansha
14898	Viraat	2021	150	Crime	8.9	294	Narendra Singh	Kritika Sachdeva	Ravi Jhanghu	Lokesh Tilakdhari
15071	Win Marathon	2018	94	Drama, Sport	9.0	70	Aryan Neeraaj Anand	Shubhanshu Satyadeo	Aryan Neeraaj Anand	Ankita Bahuguna
15288	Yeh Suhaagraat Impossible	2019	92	Comedy	8.6	500	Abhinav Thakur	Preetika Chauhan	Aarav Mavi	Aloknath Pathak

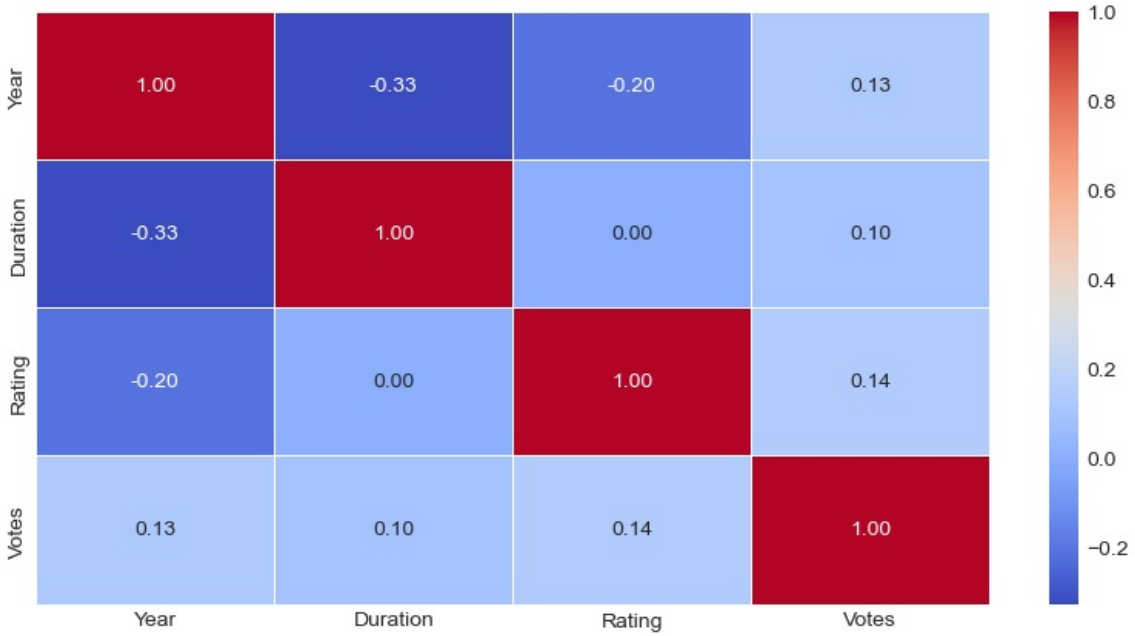
In [27]:
```
rating_counts = high_rating_movies.groupby('Rating')['Name'].count().reset_index()
rating_counts.columns = ['Rating', 'Number of Movies']
rating_counts = rating_counts.sort_values(by='Rating', ascending=False)
rating_counts
```

Out[27]:

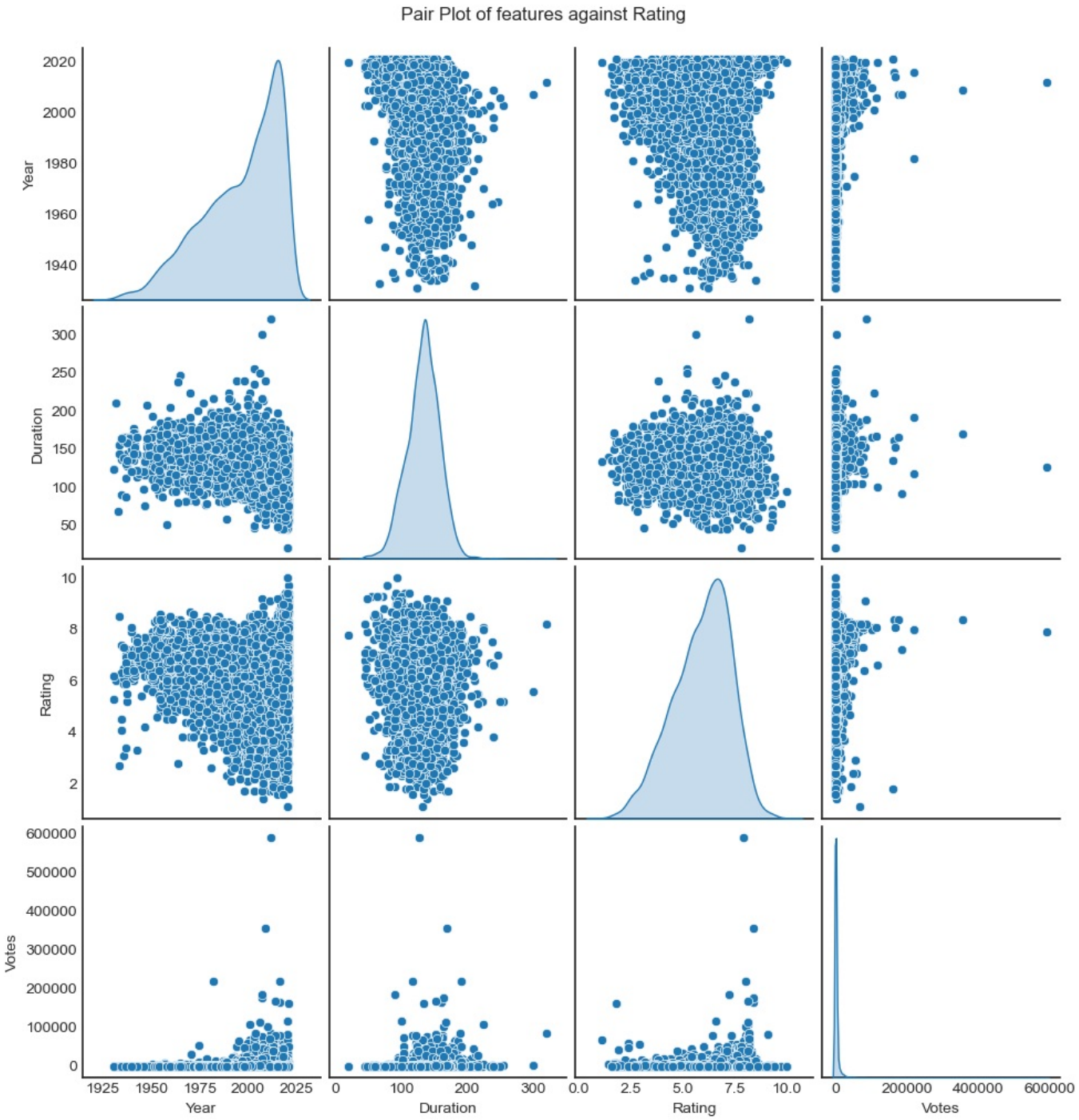
	Rating	Number of Movies
10	10.0	1
9	9.7	1
8	9.4	3
7	9.3	7
6	9.2	2
5	9.1	4
4	9.0	5
3	8.9	5
2	8.8	6
1	8.7	8
0	8.6	16

In [28]:
```
new_df=  df.drop(columns=['Name','Actor 1','Actor 2','Actor 3','Director','Genre'])
corr=new_df.corr()
plt.figure(figsize=(10,5))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", linewidth=0.5)
```

Out[28]: <Axes: >



In [29]:
```
sns.pairplot(new_df,diag_kind='kde')
plt.suptitle(' Pair Plot of features against Rating', y=1.02)
plt.show()
```



In [30]:
```
df= df.drop('Name',axis=1)
```

In [31]:
```
Genre_Average_Rating = df.groupby('Genre')['Rating'].transform('mean')
df['Genre_Average_Rating'] =Genre_Average_Rating
Director_Average_Rating = df.groupby('Director')['Rating'].transform('mean')
df['Director_Average_Rating'] =Director_Average_Rating
Actor1_Average_Rating = df.groupby('Actor 1')['Rating'].transform('mean')
df['Actor1_Average_Rating'] =Actor1_Average_Rating
Actor2_Average_Rating = df.groupby('Actor 2')['Rating'].transform('mean')
df['Actor2_Average_Rating'] =Actor2_Average_Rating
Actor3_Average_Rating = df.groupby('Actor 3')['Rating'].transform('mean')
df['Actor3_Average_Rating'] =Actor3_Average_Rating
df
```

Out[31]:

	Year	Duration	Genre	Rating	Votes	Director	Actor 1	Actor 2	Actor 3	Genre_Average_Rating	Director_
1	2019	109	Drama	7.0	8	Gaurav Bakshi	Rasika Dugal	Vivek Ghamande	Arvind Jangid	6.415521	
3	2019	110	Comedy, Romance	4.4	35	Ovais Khan	Prateik	Ishita Raj	Siddhant Kapoor	5.716822	
5	1997	147	Comedy, Drama, Musical	4.7	827	Rahul Rawail	Bobby Deol	Aishwarya Rai Bachchan	Shammi Kapoor	6.242222	
6	2005	142	Drama, Romance, War	7.4	1086	Shoojit Sircar	Jimmy Sheirgill	Minissha Lamba	Yashpal Sharma	6.820000	
8	2012	82	Horror, Mystery, Thriller	5.6	326	Allyson Patel	Yash Dave	Muntazir Ahmad	Kiran Bhatia	5.477778	
...	...	...	...	...	...	...	...	...	...	...	...
15493	2015	115	Drama	6.1	408	Mozez Singh	Vicky Kaushal	Sarah Jane Dias	Raaghavv Chanana	6.415521	
15494	2001	153	Biography, Drama, History	6.2	1496	Shyam Benegal	Karisma Kapoor	Rekha	Manoj Bajpayee	6.950000	
15503	1989	125	Action, Crime, Drama	5.8	44	S.P. Muthuraman	Chiranjeevi	Jayamalini	Rajinikanth	5.743465	
15505	1999	129	Action, Drama	4.5	655	Kuku Kohli	Akshay Kumar	Twinkle Khanna	Aruna Irani	5.516505	
15508	1998	130	Action, Drama	6.2	20	K.C. Bokadia	Dharmendra	Jaya Prada	Arjun Sarja	5.516505	

5659 rows × 14 columns

In [32]:
```
from sklearn.model_selection import train_test_split
x= df[['Year','Votes','Duration','Genre_Average_Rating','Director_Average_Rating','Actor1_Average_Rating','Acto
y=df['Rating']
```

In [33]:
```
x_train,x_test,y_train,y_test = train_test_split(x,y, test_size=0.2)
```

In [34]:
```
train_data= x_train.join(y_train)
train_data
```

Out[34]:

	Year	Votes	Duration	Genre_Average_Rating	Director_Average_Rating	Actor1_Average_Rating	Actor2_Average_Rating	Ac
566	1989	28	168	5.477966	4.133333	5.594340	5.025000	
4187	2019	6418	147	5.337778	5.000000	6.875000	4.150000	
13032	2020	1322	129	5.716822	6.270000	5.912500	6.300000	
7495	2005	245	123	5.934483	4.300000	3.700000	5.823529	
11764	2013	426	144	5.516505	6.296552	4.980000	5.100000	
...	...	...	...	...	...	...	...	...
2441	2019	43	152	5.100000	5.400000	5.926667	5.400000	
5584	1991	17	129	4.648649	4.300000	5.233333	4.100000	
13266	2007	8	160	5.522222	4.950000	6.045455	5.272222	
14532	2002	2787	160	5.390000	5.830000	4.200000	6.350000	
12650	2020	25	104	5.295238	3.450000	2.500000	2.500000	

4527 rows × 9 columns

In [35]:
```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor, AdaBoostRegressor, StackingRegres
from xgboost import XGBRegressor
from catboost import CatBoostRegressor
import lightgbm as lgb
from ngboost import NGBRegressor

# Assuming train_data and test_data are already defined

x_train, y_train = train_data[['Year','Votes','Duration','Genre_Average_Rating','Director_Average_Rating','Acto

# Linear Regression
```

```
fitted_model_lr = LinearRegression()
fitted_model_lr.fit(x_train,y_train)
y_pred_lr = fitted_model_lr.predict(x_test)

# Random Forest
fitted_model_rf = RandomForestRegressor()
fitted_model_rf.fit(x_train, y_train)
y_pred_rf = fitted_model_rf.predict(x_test)

# Gradient Boosting
fitted_model_gb = GradientBoostingRegressor()
fitted_model_gb.fit(x_train, y_train)
y_pred_gb = fitted_model_gb.predict(x_test)

# XGBoost
fitted_model_xgb = XGBRegressor()
fitted_model_xgb.fit(x_train, y_train)
y_pred_xgb = fitted_model_xgb.predict(x_test)

# CatBoost
fitted_model_cb = CatBoostRegressor()
fitted_model_cb.fit(x_train, y_train)
y_pred_cb = fitted_model_cb.predict(x_test)

# LightGBM
fitted_model_lgb = lgb.LGBMRegressor()
fitted_model_lgb.fit(x_train, y_train)
y_pred_lgb = fitted_model_lgb.predict(x_test)

# AdaBoost
fitted_model_ab = AdaBoostRegressor()
fitted_model_ab.fit(x_train, y_train)
y_pred_ab = fitted_model_ab.predict(x_test)

# NGBoost
fitted_model_ngb = NGBRegressor()
fitted_model_ngb.fit(x_train, y_train)
y_pred_ngb = fitted_model_ngb.predict(x_test)

# Stacking Regressor
estimators = [('lr', LinearRegression()), ('rf', RandomForestRegressor()), ('gb', GradientBoostingRegressor())]
fitted_model_sr = StackingRegressor(estimators=estimators, final_estimator=LinearRegression())
fitted_model_sr.fit(x_train, y_train)
y_pred_sr = fitted_model_sr.predict(x_test)
```

```
Learning rate set to 0.051975
0:      learn: 1.3197095        total: 169ms    remaining: 2m 48s
1:      learn: 1.2785944        total: 176ms    remaining: 1m 27s
2:      learn: 1.2397246        total: 186ms    remaining: 1m 1s
3:      learn: 1.2006806        total: 193ms    remaining: 48s
4:      learn: 1.1634788        total: 200ms    remaining: 39.7s
5:      learn: 1.1283124        total: 208ms    remaining: 34.5s
6:      learn: 1.0955609        total: 216ms    remaining: 30.6s
7:      learn: 1.0658727        total: 222ms    remaining: 27.6s
8:      learn: 1.0368229        total: 229ms    remaining: 25.2s
9:      learn: 1.0097532        total: 237ms    remaining: 23.4s
10:     learn: 0.9841605        total: 243ms    remaining: 21.9s
11:     learn: 0.9610697        total: 252ms    remaining: 20.8s
12:     learn: 0.9388528        total: 259ms    remaining: 19.7s
13:     learn: 0.9179836        total: 266ms    remaining: 18.7s
14:     learn: 0.8981150        total: 276ms    remaining: 18.1s
15:     learn: 0.8794107        total: 282ms    remaining: 17.3s
16:     learn: 0.8623191        total: 287ms    remaining: 16.6s
17:     learn: 0.8460505        total: 292ms    remaining: 16s
18:     learn: 0.8316336        total: 298ms    remaining: 15.4s
19:     learn: 0.8187387        total: 304ms    remaining: 14.9s
20:     learn: 0.8044574        total: 309ms    remaining: 14.4s
21:     learn: 0.7921773        total: 314ms    remaining: 13.9s
22:     learn: 0.7801939        total: 319ms    remaining: 13.5s
23:     learn: 0.7684751        total: 324ms    remaining: 13.2s
24:     learn: 0.7579933        total: 329ms    remaining: 12.8s
25:     learn: 0.7482042        total: 335ms    remaining: 12.6s
26:     learn: 0.7384631        total: 341ms    remaining: 12.3s
27:     learn: 0.7298833        total: 347ms    remaining: 12s
28:     learn: 0.7215664        total: 352ms    remaining: 11.8s
29:     learn: 0.7140566        total: 357ms    remaining: 11.5s
30:     learn: 0.7070990        total: 362ms    remaining: 11.3s
31:     learn: 0.7005311        total: 367ms    remaining: 11.1s
32:     learn: 0.6941430        total: 372ms    remaining: 10.9s
33:     learn: 0.6882364        total: 377ms    remaining: 10.7s
34:     learn: 0.6823287        total: 382ms    remaining: 10.5s
35:     learn: 0.6773211        total: 389ms    remaining: 10.4s
36:     learn: 0.6718084        total: 394ms    remaining: 10.2s
37:     learn: 0.6671013        total: 399ms    remaining: 10.1s
```

```
38:     learn: 0.6618483        total: 404ms    remaining: 9.94s
39:     learn: 0.6581708        total: 408ms    remaining: 9.8s
40:     learn: 0.6540960        total: 413ms    remaining: 9.65s
41:     learn: 0.6506121        total: 418ms    remaining: 9.53s
42:     learn: 0.6471572        total: 423ms    remaining: 9.4s
43:     learn: 0.6442478        total: 427ms    remaining: 9.29s
44:     learn: 0.6406543        total: 432ms    remaining: 9.18s
45:     learn: 0.6373966        total: 438ms    remaining: 9.07s
46:     learn: 0.6341112        total: 442ms    remaining: 8.97s
47:     learn: 0.6314902        total: 447ms    remaining: 8.87s
48:     learn: 0.6290556        total: 452ms    remaining: 8.77s
49:     learn: 0.6258267        total: 457ms    remaining: 8.68s
50:     learn: 0.6236068        total: 461ms    remaining: 8.59s
51:     learn: 0.6213895        total: 466ms    remaining: 8.5s
52:     learn: 0.6184344        total: 471ms    remaining: 8.42s
53:     learn: 0.6160751        total: 476ms    remaining: 8.34s
54:     learn: 0.6139414        total: 481ms    remaining: 8.27s
55:     learn: 0.6120079        total: 488ms    remaining: 8.23s
56:     learn: 0.6102385        total: 494ms    remaining: 8.17s
57:     learn: 0.6077800        total: 499ms    remaining: 8.11s
58:     learn: 0.6060714        total: 504ms    remaining: 8.04s
59:     learn: 0.6035058        total: 510ms    remaining: 7.98s
60:     learn: 0.6016013        total: 515ms    remaining: 7.92s
61:     learn: 0.6003192        total: 519ms    remaining: 7.85s
62:     learn: 0.5981975        total: 524ms    remaining: 7.8s
63:     learn: 0.5968164        total: 530ms    remaining: 7.76s
64:     learn: 0.5953661        total: 535ms    remaining: 7.7s
65:     learn: 0.5934476        total: 540ms    remaining: 7.64s
66:     learn: 0.5915473        total: 545ms    remaining: 7.59s
67:     learn: 0.5904119        total: 552ms    remaining: 7.56s
68:     learn: 0.5893000        total: 557ms    remaining: 7.52s
69:     learn: 0.5884120        total: 563ms    remaining: 7.48s
70:     learn: 0.5874488        total: 569ms    remaining: 7.45s
71:     learn: 0.5865773        total: 575ms    remaining: 7.41s
72:     learn: 0.5856489        total: 581ms    remaining: 7.38s
73:     learn: 0.5849723        total: 587ms    remaining: 7.35s
74:     learn: 0.5833913        total: 595ms    remaining: 7.34s
75:     learn: 0.5816881        total: 601ms    remaining: 7.3s
76:     learn: 0.5802638        total: 607ms    remaining: 7.28s
77:     learn: 0.5788212        total: 612ms    remaining: 7.24s
78:     learn: 0.5778517        total: 617ms    remaining: 7.19s
79:     learn: 0.5764756        total: 623ms    remaining: 7.17s
80:     learn: 0.5754073        total: 628ms    remaining: 7.13s
81:     learn: 0.5746227        total: 634ms    remaining: 7.1s
82:     learn: 0.5738556        total: 640ms    remaining: 7.08s
83:     learn: 0.5732171        total: 646ms    remaining: 7.05s
84:     learn: 0.5725781        total: 651ms    remaining: 7.01s
85:     learn: 0.5716547        total: 656ms    remaining: 6.98s
86:     learn: 0.5706210        total: 662ms    remaining: 6.94s
87:     learn: 0.5697948        total: 668ms    remaining: 6.92s
88:     learn: 0.5686093        total: 674ms    remaining: 6.9s
89:     learn: 0.5676451        total: 680ms    remaining: 6.87s
90:     learn: 0.5669837        total: 685ms    remaining: 6.84s
91:     learn: 0.5664947        total: 690ms    remaining: 6.81s
92:     learn: 0.5657002        total: 696ms    remaining: 6.79s
93:     learn: 0.5650255        total: 702ms    remaining: 6.77s
94:     learn: 0.5636899        total: 708ms    remaining: 6.74s
95:     learn: 0.5629680        total: 713ms    remaining: 6.71s
96:     learn: 0.5621764        total: 719ms    remaining: 6.69s
97:     learn: 0.5617663        total: 725ms    remaining: 6.67s
98:     learn: 0.5613475        total: 731ms    remaining: 6.65s
99:     learn: 0.5609152        total: 736ms    remaining: 6.63s
100:    learn: 0.5604801        total: 741ms    remaining: 6.59s
101:    learn: 0.5596961        total: 746ms    remaining: 6.57s
102:    learn: 0.5587217        total: 751ms    remaining: 6.54s
103:    learn: 0.5582901        total: 758ms    remaining: 6.53s
104:    learn: 0.5579387        total: 764ms    remaining: 6.51s
105:    learn: 0.5576059        total: 770ms    remaining: 6.49s
106:    learn: 0.5572230        total: 775ms    remaining: 6.47s
107:    learn: 0.5567299        total: 780ms    remaining: 6.44s
108:    learn: 0.5561103        total: 785ms    remaining: 6.41s
109:    learn: 0.5552296        total: 791ms    remaining: 6.4s
110:    learn: 0.5547114        total: 797ms    remaining: 6.38s
111:    learn: 0.5544267        total: 802ms    remaining: 6.36s
112:    learn: 0.5539398        total: 808ms    remaining: 6.34s
113:    learn: 0.5535984        total: 814ms    remaining: 6.33s
114:    learn: 0.5526156        total: 819ms    remaining: 6.3s
115:    learn: 0.5522391        total: 824ms    remaining: 6.28s
116:    learn: 0.5515336        total: 830ms    remaining: 6.26s
117:    learn: 0.5510009        total: 836ms    remaining: 6.25s
118:    learn: 0.5503772        total: 842ms    remaining: 6.23s
119:    learn: 0.5500577        total: 848ms    remaining: 6.22s
120:    learn: 0.5496897        total: 853ms    remaining: 6.2s
```

```
121:    learn: 0.5489681        total: 859ms    remaining: 6.18s
122:    learn: 0.5487347        total: 864ms    remaining: 6.16s
123:    learn: 0.5484789        total: 870ms    remaining: 6.14s
124:    learn: 0.5481364        total: 877ms    remaining: 6.13s
125:    learn: 0.5478781        total: 883ms    remaining: 6.12s
126:    learn: 0.5473935        total: 889ms    remaining: 6.11s
127:    learn: 0.5470807        total: 895ms    remaining: 6.1s
128:    learn: 0.5467341        total: 901ms    remaining: 6.08s
129:    learn: 0.5461344        total: 908ms    remaining: 6.07s
130:    learn: 0.5458786        total: 915ms    remaining: 6.07s
131:    learn: 0.5456084        total: 922ms    remaining: 6.06s
132:    learn: 0.5452692        total: 928ms    remaining: 6.05s
133:    learn: 0.5448995        total: 935ms    remaining: 6.04s
134:    learn: 0.5442987        total: 942ms    remaining: 6.03s
135:    learn: 0.5435671        total: 948ms    remaining: 6.02s
136:    learn: 0.5434397        total: 954ms    remaining: 6.01s
137:    learn: 0.5431077        total: 960ms    remaining: 6s
138:    learn: 0.5428023        total: 965ms    remaining: 5.98s
139:    learn: 0.5421723        total: 970ms    remaining: 5.96s
140:    learn: 0.5416977        total: 975ms    remaining: 5.94s
141:    learn: 0.5409560        total: 981ms    remaining: 5.93s
142:    learn: 0.5406001        total: 986ms    remaining: 5.91s
143:    learn: 0.5405300        total: 990ms    remaining: 5.88s
144:    learn: 0.5399679        total: 997ms    remaining: 5.88s
145:    learn: 0.5395085        total: 1s       remaining: 5.86s
146:    learn: 0.5391984        total: 1.01s    remaining: 5.84s
147:    learn: 0.5387227        total: 1.01s    remaining: 5.84s
148:    learn: 0.5384090        total: 1.02s    remaining: 5.82s
149:    learn: 0.5381950        total: 1.02s    remaining: 5.81s
150:    learn: 0.5375401        total: 1.03s    remaining: 5.79s
151:    learn: 0.5372982        total: 1.04s    remaining: 5.78s
152:    learn: 0.5367944        total: 1.04s    remaining: 5.77s
153:    learn: 0.5362174        total: 1.05s    remaining: 5.75s
154:    learn: 0.5358048        total: 1.05s    remaining: 5.74s
155:    learn: 0.5355645        total: 1.06s    remaining: 5.72s
156:    learn: 0.5351562        total: 1.06s    remaining: 5.71s
157:    learn: 0.5345225        total: 1.07s    remaining: 5.7s
158:    learn: 0.5341142        total: 1.07s    remaining: 5.68s
159:    learn: 0.5335613        total: 1.08s    remaining: 5.67s
160:    learn: 0.5332078        total: 1.08s    remaining: 5.65s
161:    learn: 0.5330388        total: 1.09s    remaining: 5.64s
162:    learn: 0.5323578        total: 1.09s    remaining: 5.63s
163:    learn: 0.5321728        total: 1.1s     remaining: 5.61s
164:    learn: 0.5317882        total: 1.11s    remaining: 5.6s
165:    learn: 0.5313760        total: 1.11s    remaining: 5.59s
166:    learn: 0.5310235        total: 1.12s    remaining: 5.57s
167:    learn: 0.5307005        total: 1.12s    remaining: 5.56s
168:    learn: 0.5302257        total: 1.13s    remaining: 5.55s
169:    learn: 0.5298227        total: 1.14s    remaining: 5.54s
170:    learn: 0.5293462        total: 1.14s    remaining: 5.53s
171:    learn: 0.5291284        total: 1.15s    remaining: 5.51s
172:    learn: 0.5288606        total: 1.15s    remaining: 5.5s
173:    learn: 0.5282321        total: 1.16s    remaining: 5.49s
174:    learn: 0.5280050        total: 1.16s    remaining: 5.47s
175:    learn: 0.5274836        total: 1.17s    remaining: 5.45s
176:    learn: 0.5270947        total: 1.17s    remaining: 5.44s
177:    learn: 0.5267130        total: 1.17s    remaining: 5.42s
178:    learn: 0.5264482        total: 1.18s    remaining: 5.41s
179:    learn: 0.5259727        total: 1.18s    remaining: 5.39s
180:    learn: 0.5256156        total: 1.19s    remaining: 5.38s
181:    learn: 0.5254207        total: 1.2s     remaining: 5.38s
182:    learn: 0.5253325        total: 1.2s     remaining: 5.37s
183:    learn: 0.5248987        total: 1.21s    remaining: 5.36s
184:    learn: 0.5243065        total: 1.21s    remaining: 5.35s
185:    learn: 0.5241564        total: 1.22s    remaining: 5.34s
186:    learn: 0.5238641        total: 1.23s    remaining: 5.33s
187:    learn: 0.5236051        total: 1.23s    remaining: 5.32s
188:    learn: 0.5232535        total: 1.24s    remaining: 5.3s
189:    learn: 0.5229843        total: 1.24s    remaining: 5.29s
190:    learn: 0.5228241        total: 1.25s    remaining: 5.28s
191:    learn: 0.5221829        total: 1.25s    remaining: 5.27s
192:    learn: 0.5218108        total: 1.26s    remaining: 5.26s
193:    learn: 0.5216212        total: 1.26s    remaining: 5.25s
194:    learn: 0.5213686        total: 1.27s    remaining: 5.24s
195:    learn: 0.5211917        total: 1.27s    remaining: 5.23s
196:    learn: 0.5208148        total: 1.28s    remaining: 5.22s
197:    learn: 0.5202619        total: 1.28s    remaining: 5.21s
198:    learn: 0.5199900        total: 1.29s    remaining: 5.2s
199:    learn: 0.5197616        total: 1.3s     remaining: 5.18s
200:    learn: 0.5196065        total: 1.3s     remaining: 5.17s
201:    learn: 0.5193919        total: 1.31s    remaining: 5.16s
202:    learn: 0.5190142        total: 1.31s    remaining: 5.15s
203:    learn: 0.5184087        total: 1.32s    remaining: 5.14s
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204:    learn: 0.5181608        total: 1.32s    remaining: 5.13s
205:    learn: 0.5179727        total: 1.33s    remaining: 5.11s
206:    learn: 0.5174485        total: 1.33s    remaining: 5.1s
207:    learn: 0.5172441        total: 1.34s    remaining: 5.09s
208:    learn: 0.5168341        total: 1.34s    remaining: 5.07s
209:    learn: 0.5166327        total: 1.34s    remaining: 5.06s
210:    learn: 0.5163286        total: 1.35s    remaining: 5.05s
211:    learn: 0.5161273        total: 1.35s    remaining: 5.03s
212:    learn: 0.5155905        total: 1.36s    remaining: 5.02s
213:    learn: 0.5155192        total: 1.36s    remaining: 5.01s
214:    learn: 0.5149885        total: 1.37s    remaining: 5s
215:    learn: 0.5146199        total: 1.37s    remaining: 4.99s
216:    learn: 0.5144841        total: 1.38s    remaining: 4.97s
217:    learn: 0.5141914        total: 1.38s    remaining: 4.96s
218:    learn: 0.5137875        total: 1.39s    remaining: 4.95s
219:    learn: 0.5135409        total: 1.39s    remaining: 4.94s
220:    learn: 0.5133292        total: 1.4s     remaining: 4.93s
221:    learn: 0.5129064        total: 1.4s     remaining: 4.92s
222:    learn: 0.5125136        total: 1.41s    remaining: 4.91s
223:    learn: 0.5120318        total: 1.42s    remaining: 4.9s
224:    learn: 0.5117608        total: 1.42s    remaining: 4.89s
225:    learn: 0.5116904        total: 1.43s    remaining: 4.88s
226:    learn: 0.5114832        total: 1.43s    remaining: 4.87s
227:    learn: 0.5113397        total: 1.43s    remaining: 4.86s
228:    learn: 0.5109395        total: 1.44s    remaining: 4.84s
229:    learn: 0.5104068        total: 1.44s    remaining: 4.83s
230:    learn: 0.5100084        total: 1.45s    remaining: 4.82s
231:    learn: 0.5091752        total: 1.45s    remaining: 4.81s
232:    learn: 0.5086407        total: 1.46s    remaining: 4.8s
233:    learn: 0.5080939        total: 1.46s    remaining: 4.79s
234:    learn: 0.5079729        total: 1.47s    remaining: 4.78s
235:    learn: 0.5078210        total: 1.47s    remaining: 4.77s
236:    learn: 0.5072175        total: 1.48s    remaining: 4.76s
237:    learn: 0.5069456        total: 1.48s    remaining: 4.75s
238:    learn: 0.5067324        total: 1.49s    remaining: 4.74s
239:    learn: 0.5065237        total: 1.49s    remaining: 4.72s
240:    learn: 0.5062911        total: 1.5s     remaining: 4.71s
241:    learn: 0.5062536        total: 1.5s     remaining: 4.7s
242:    learn: 0.5059859        total: 1.5s     remaining: 4.69s
243:    learn: 0.5056766        total: 1.51s    remaining: 4.68s
244:    learn: 0.5047545        total: 1.51s    remaining: 4.67s
245:    learn: 0.5042730        total: 1.52s    remaining: 4.66s
246:    learn: 0.5037297        total: 1.52s    remaining: 4.65s
247:    learn: 0.5033147        total: 1.53s    remaining: 4.64s
248:    learn: 0.5030481        total: 1.53s    remaining: 4.63s
249:    learn: 0.5025424        total: 1.54s    remaining: 4.62s
250:    learn: 0.5018922        total: 1.54s    remaining: 4.61s
251:    learn: 0.5016394        total: 1.55s    remaining: 4.6s
252:    learn: 0.5014397        total: 1.55s    remaining: 4.59s
253:    learn: 0.5008794        total: 1.56s    remaining: 4.58s
254:    learn: 0.5007397        total: 1.56s    remaining: 4.57s
255:    learn: 0.5005139        total: 1.57s    remaining: 4.56s
256:    learn: 0.5002828        total: 1.57s    remaining: 4.55s
257:    learn: 0.4999798        total: 1.58s    remaining: 4.54s
258:    learn: 0.4996831        total: 1.58s    remaining: 4.53s
259:    learn: 0.4995110        total: 1.59s    remaining: 4.52s
260:    learn: 0.4993154        total: 1.59s    remaining: 4.51s
261:    learn: 0.4989342        total: 1.6s     remaining: 4.5s
262:    learn: 0.4987627        total: 1.6s     remaining: 4.49s
263:    learn: 0.4982045        total: 1.61s    remaining: 4.49s
264:    learn: 0.4979521        total: 1.61s    remaining: 4.48s
265:    learn: 0.4978284        total: 1.62s    remaining: 4.47s
266:    learn: 0.4977184        total: 1.62s    remaining: 4.46s
267:    learn: 0.4975416        total: 1.63s    remaining: 4.45s
268:    learn: 0.4974986        total: 1.63s    remaining: 4.44s
269:    learn: 0.4967503        total: 1.64s    remaining: 4.43s
270:    learn: 0.4966758        total: 1.64s    remaining: 4.42s
271:    learn: 0.4964765        total: 1.65s    remaining: 4.41s
272:    learn: 0.4960711        total: 1.65s    remaining: 4.4s
273:    learn: 0.4956381        total: 1.66s    remaining: 4.39s
274:    learn: 0.4956156        total: 1.66s    remaining: 4.38s
275:    learn: 0.4951707        total: 1.67s    remaining: 4.38s
276:    learn: 0.4945720        total: 1.67s    remaining: 4.37s
277:    learn: 0.4945070        total: 1.68s    remaining: 4.36s
278:    learn: 0.4944566        total: 1.68s    remaining: 4.35s
279:    learn: 0.4943307        total: 1.69s    remaining: 4.35s
280:    learn: 0.4940637        total: 1.7s     remaining: 4.34s
281:    learn: 0.4939667        total: 1.7s     remaining: 4.33s
282:    learn: 0.4936086        total: 1.71s    remaining: 4.32s
283:    learn: 0.4932770        total: 1.71s    remaining: 4.31s
284:    learn: 0.4929893        total: 1.72s    remaining: 4.3s
285:    learn: 0.4928664        total: 1.72s    remaining: 4.29s
286:    learn: 0.4928101        total: 1.73s    remaining: 4.29s
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287:    learn: 0.4924507        total: 1.73s    remaining: 4.28s
288:    learn: 0.4923545        total: 1.73s    remaining: 4.27s
289:    learn: 0.4923185        total: 1.74s    remaining: 4.26s
290:    learn: 0.4920038        total: 1.74s    remaining: 4.25s
291:    learn: 0.4919845        total: 1.75s    remaining: 4.24s
292:    learn: 0.4914977        total: 1.75s    remaining: 4.23s
293:    learn: 0.4911820        total: 1.76s    remaining: 4.22s
294:    learn: 0.4910146        total: 1.76s    remaining: 4.22s
295:    learn: 0.4905772        total: 1.77s    remaining: 4.21s
296:    learn: 0.4902552        total: 1.77s    remaining: 4.2s
297:    learn: 0.4899483        total: 1.78s    remaining: 4.19s
298:    learn: 0.4897027        total: 1.78s    remaining: 4.18s
299:    learn: 0.4896097        total: 1.79s    remaining: 4.18s
300:    learn: 0.4891995        total: 1.79s    remaining: 4.17s
301:    learn: 0.4886812        total: 1.8s     remaining: 4.16s
302:    learn: 0.4885598        total: 1.8s     remaining: 4.15s
303:    learn: 0.4885133        total: 1.81s    remaining: 4.14s
304:    learn: 0.4883577        total: 1.81s    remaining: 4.14s
305:    learn: 0.4882243        total: 1.82s    remaining: 4.13s
306:    learn: 0.4879208        total: 1.82s    remaining: 4.12s
307:    learn: 0.4877420        total: 1.83s    remaining: 4.11s
308:    learn: 0.4875957        total: 1.84s    remaining: 4.11s
309:    learn: 0.4873184        total: 1.84s    remaining: 4.1s
310:    learn: 0.4870173        total: 1.85s    remaining: 4.09s
311:    learn: 0.4867397        total: 1.85s    remaining: 4.08s
312:    learn: 0.4866544        total: 1.86s    remaining: 4.08s
313:    learn: 0.4866354        total: 1.86s    remaining: 4.07s
314:    learn: 0.4864323        total: 1.87s    remaining: 4.06s
315:    learn: 0.4862348        total: 1.87s    remaining: 4.05s
316:    learn: 0.4856797        total: 1.88s    remaining: 4.05s
317:    learn: 0.4855211        total: 1.88s    remaining: 4.04s
318:    learn: 0.4851588        total: 1.89s    remaining: 4.03s
319:    learn: 0.4849016        total: 1.89s    remaining: 4.03s
320:    learn: 0.4846097        total: 1.9s     remaining: 4.02s
321:    learn: 0.4842964        total: 1.9s     remaining: 4.01s
322:    learn: 0.4838024        total: 1.91s    remaining: 4s
323:    learn: 0.4834016        total: 1.92s    remaining: 4s
324:    learn: 0.4829967        total: 1.92s    remaining: 3.99s
325:    learn: 0.4824500        total: 1.93s    remaining: 3.98s
326:    learn: 0.4824087        total: 1.93s    remaining: 3.97s
327:    learn: 0.4822213        total: 1.94s    remaining: 3.96s
328:    learn: 0.4819949        total: 1.94s    remaining: 3.96s
329:    learn: 0.4816315        total: 1.95s    remaining: 3.95s
330:    learn: 0.4813471        total: 1.95s    remaining: 3.94s
331:    learn: 0.4813131        total: 1.96s    remaining: 3.94s
332:    learn: 0.4811071        total: 1.96s    remaining: 3.93s
333:    learn: 0.4810751        total: 1.97s    remaining: 3.92s
334:    learn: 0.4807556        total: 1.97s    remaining: 3.91s
335:    learn: 0.4803533        total: 1.98s    remaining: 3.91s
336:    learn: 0.4802334        total: 1.98s    remaining: 3.9s
337:    learn: 0.4800800        total: 1.99s    remaining: 3.89s
338:    learn: 0.4796943        total: 1.99s    remaining: 3.88s
339:    learn: 0.4793733        total: 2s       remaining: 3.88s
340:    learn: 0.4788706        total: 2s       remaining: 3.87s
341:    learn: 0.4784072        total: 2.01s    remaining: 3.87s
342:    learn: 0.4780606        total: 2.01s    remaining: 3.86s
343:    learn: 0.4775149        total: 2.02s    remaining: 3.85s
344:    learn: 0.4770396        total: 2.02s    remaining: 3.84s
345:    learn: 0.4767745        total: 2.03s    remaining: 3.84s
346:    learn: 0.4766569        total: 2.03s    remaining: 3.83s
347:    learn: 0.4763441        total: 2.04s    remaining: 3.82s
348:    learn: 0.4761420        total: 2.04s    remaining: 3.81s
349:    learn: 0.4759595        total: 2.05s    remaining: 3.81s
350:    learn: 0.4756123        total: 2.06s    remaining: 3.8s
351:    learn: 0.4754094        total: 2.06s    remaining: 3.79s
352:    learn: 0.4753772        total: 2.07s    remaining: 3.79s
353:    learn: 0.4751409        total: 2.07s    remaining: 3.78s
354:    learn: 0.4748709        total: 2.08s    remaining: 3.77s
355:    learn: 0.4744846        total: 2.08s    remaining: 3.77s
356:    learn: 0.4741402        total: 2.09s    remaining: 3.76s
357:    learn: 0.4737828        total: 2.09s    remaining: 3.75s
358:    learn: 0.4736948        total: 2.1s     remaining: 3.74s
359:    learn: 0.4735087        total: 2.1s     remaining: 3.74s
360:    learn: 0.4733872        total: 2.11s    remaining: 3.73s
361:    learn: 0.4732980        total: 2.11s    remaining: 3.72s
362:    learn: 0.4729561        total: 2.12s    remaining: 3.72s
363:    learn: 0.4726706        total: 2.12s    remaining: 3.71s
364:    learn: 0.4724031        total: 2.13s    remaining: 3.7s
365:    learn: 0.4721023        total: 2.13s    remaining: 3.7s
366:    learn: 0.4720088        total: 2.14s    remaining: 3.69s
367:    learn: 0.4716494        total: 2.14s    remaining: 3.68s
368:    learn: 0.4714490        total: 2.15s    remaining: 3.67s
369:    learn: 0.4712737        total: 2.15s    remaining: 3.67s
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370:    learn: 0.4712390        total: 2.16s    remaining: 3.66s
371:    learn: 0.4709140        total: 2.17s    remaining: 3.65s
372:    learn: 0.4708700        total: 2.17s    remaining: 3.65s
373:    learn: 0.4707986        total: 2.18s    remaining: 3.64s
374:    learn: 0.4704794        total: 2.18s    remaining: 3.64s
375:    learn: 0.4702974        total: 2.19s    remaining: 3.63s
376:    learn: 0.4697997        total: 2.19s    remaining: 3.63s
377:    learn: 0.4692675        total: 2.2s     remaining: 3.62s
378:    learn: 0.4687103        total: 2.21s    remaining: 3.62s
379:    learn: 0.4683298        total: 2.21s    remaining: 3.61s
380:    learn: 0.4682989        total: 2.22s    remaining: 3.6s
381:    learn: 0.4679820        total: 2.22s    remaining: 3.6s
382:    learn: 0.4678181        total: 2.23s    remaining: 3.59s
383:    learn: 0.4677843        total: 2.23s    remaining: 3.58s
384:    learn: 0.4673964        total: 2.24s    remaining: 3.58s
385:    learn: 0.4672658        total: 2.24s    remaining: 3.57s
386:    learn: 0.4670445        total: 2.25s    remaining: 3.56s
387:    learn: 0.4668639        total: 2.25s    remaining: 3.56s
388:    learn: 0.4667189        total: 2.26s    remaining: 3.55s
389:    learn: 0.4662421        total: 2.26s    remaining: 3.54s
390:    learn: 0.4659244        total: 2.27s    remaining: 3.54s
391:    learn: 0.4658996        total: 2.28s    remaining: 3.53s
392:    learn: 0.4655705        total: 2.28s    remaining: 3.52s
393:    learn: 0.4652492        total: 2.29s    remaining: 3.52s
394:    learn: 0.4649491        total: 2.29s    remaining: 3.51s
395:    learn: 0.4647607        total: 2.3s     remaining: 3.5s
396:    learn: 0.4645402        total: 2.3s     remaining: 3.5s
397:    learn: 0.4643851        total: 2.31s    remaining: 3.49s
398:    learn: 0.4639769        total: 2.31s    remaining: 3.48s
399:    learn: 0.4639523        total: 2.32s    remaining: 3.48s
400:    learn: 0.4636622        total: 2.32s    remaining: 3.47s
401:    learn: 0.4633160        total: 2.33s    remaining: 3.46s
402:    learn: 0.4630808        total: 2.33s    remaining: 3.46s
403:    learn: 0.4628376        total: 2.34s    remaining: 3.45s
404:    learn: 0.4625454        total: 2.34s    remaining: 3.44s
405:    learn: 0.4623360        total: 2.35s    remaining: 3.44s
406:    learn: 0.4622241        total: 2.35s    remaining: 3.43s
407:    learn: 0.4620439        total: 2.36s    remaining: 3.42s
408:    learn: 0.4619113        total: 2.37s    remaining: 3.42s
409:    learn: 0.4619011        total: 2.37s    remaining: 3.41s
410:    learn: 0.4617332        total: 2.38s    remaining: 3.4s
411:    learn: 0.4617094        total: 2.38s    remaining: 3.4s
412:    learn: 0.4616055        total: 2.39s    remaining: 3.39s
413:    learn: 0.4612706        total: 2.39s    remaining: 3.38s
414:    learn: 0.4609410        total: 2.4s     remaining: 3.38s
415:    learn: 0.4605724        total: 2.4s     remaining: 3.37s
416:    learn: 0.4601358        total: 2.41s    remaining: 3.37s
417:    learn: 0.4598226        total: 2.41s    remaining: 3.36s
418:    learn: 0.4596619        total: 2.42s    remaining: 3.35s
419:    learn: 0.4593714        total: 2.42s    remaining: 3.35s
420:    learn: 0.4591364        total: 2.43s    remaining: 3.34s
421:    learn: 0.4587708        total: 2.43s    remaining: 3.33s
422:    learn: 0.4587412        total: 2.44s    remaining: 3.33s
423:    learn: 0.4584663        total: 2.44s    remaining: 3.32s
424:    learn: 0.4582779        total: 2.45s    remaining: 3.31s
425:    learn: 0.4582146        total: 2.45s    remaining: 3.31s
426:    learn: 0.4580469        total: 2.46s    remaining: 3.3s
427:    learn: 0.4576299        total: 2.46s    remaining: 3.29s
428:    learn: 0.4576187        total: 2.47s    remaining: 3.29s
429:    learn: 0.4573458        total: 2.47s    remaining: 3.28s
430:    learn: 0.4570712        total: 2.48s    remaining: 3.27s
431:    learn: 0.4570466        total: 2.48s    remaining: 3.27s
432:    learn: 0.4568263        total: 2.49s    remaining: 3.26s
433:    learn: 0.4568025        total: 2.49s    remaining: 3.25s
434:    learn: 0.4564843        total: 2.5s     remaining: 3.25s
435:    learn: 0.4560981        total: 2.5s     remaining: 3.24s
436:    learn: 0.4558730        total: 2.51s    remaining: 3.23s
437:    learn: 0.4556412        total: 2.52s    remaining: 3.23s
438:    learn: 0.4552394        total: 2.52s    remaining: 3.22s
439:    learn: 0.4552250        total: 2.52s    remaining: 3.21s
440:    learn: 0.4549767        total: 2.53s    remaining: 3.21s
441:    learn: 0.4548212        total: 2.54s    remaining: 3.2s
442:    learn: 0.4545802        total: 2.54s    remaining: 3.19s
443:    learn: 0.4543623        total: 2.55s    remaining: 3.19s
444:    learn: 0.4543500        total: 2.55s    remaining: 3.18s
445:    learn: 0.4540597        total: 2.56s    remaining: 3.17s
446:    learn: 0.4537562        total: 2.56s    remaining: 3.17s
447:    learn: 0.4534741        total: 2.57s    remaining: 3.16s
448:    learn: 0.4530953        total: 2.57s    remaining: 3.16s
449:    learn: 0.4528236        total: 2.58s    remaining: 3.15s
450:    learn: 0.4527448        total: 2.58s    remaining: 3.15s
451:    learn: 0.4525620        total: 2.59s    remaining: 3.14s
452:    learn: 0.4521233        total: 2.59s    remaining: 3.13s
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453:    learn: 0.4520238        total: 2.6s     remaining: 3.13s
454:    learn: 0.4518635        total: 2.6s     remaining: 3.12s
455:    learn: 0.4517867        total: 2.61s    remaining: 3.11s
456:    learn: 0.4517418        total: 2.62s    remaining: 3.11s
457:    learn: 0.4513501        total: 2.62s    remaining: 3.1s
458:    learn: 0.4511697        total: 2.63s    remaining: 3.1s
459:    learn: 0.4509471        total: 2.63s    remaining: 3.09s
460:    learn: 0.4506953        total: 2.64s    remaining: 3.08s
461:    learn: 0.4506675        total: 2.64s    remaining: 3.08s
462:    learn: 0.4501962        total: 2.65s    remaining: 3.07s
463:    learn: 0.4501261        total: 2.65s    remaining: 3.06s
464:    learn: 0.4499607        total: 2.66s    remaining: 3.06s
465:    learn: 0.4495814        total: 2.66s    remaining: 3.05s
466:    learn: 0.4491757        total: 2.67s    remaining: 3.05s
467:    learn: 0.4487740        total: 2.68s    remaining: 3.04s
468:    learn: 0.4484972        total: 2.68s    remaining: 3.04s
469:    learn: 0.4483334        total: 2.69s    remaining: 3.03s
470:    learn: 0.4481821        total: 2.69s    remaining: 3.02s
471:    learn: 0.4479782        total: 2.7s     remaining: 3.02s
472:    learn: 0.4478365        total: 2.7s     remaining: 3.01s
473:    learn: 0.4474418        total: 2.71s    remaining: 3.01s
474:    learn: 0.4472534        total: 2.71s    remaining: 3s
475:    learn: 0.4469969        total: 2.72s    remaining: 2.99s
476:    learn: 0.4468176        total: 2.73s    remaining: 2.99s
477:    learn: 0.4465886        total: 2.73s    remaining: 2.98s
478:    learn: 0.4463299        total: 2.74s    remaining: 2.98s
479:    learn: 0.4460458        total: 2.74s    remaining: 2.97s
480:    learn: 0.4458607        total: 2.75s    remaining: 2.96s
481:    learn: 0.4455542        total: 2.75s    remaining: 2.96s
482:    learn: 0.4453137        total: 2.76s    remaining: 2.95s
483:    learn: 0.4451618        total: 2.76s    remaining: 2.94s
484:    learn: 0.4449655        total: 2.77s    remaining: 2.94s
485:    learn: 0.4447200        total: 2.77s    remaining: 2.93s
486:    learn: 0.4443649        total: 2.78s    remaining: 2.93s
487:    learn: 0.4443435        total: 2.78s    remaining: 2.92s
488:    learn: 0.4441633        total: 2.79s    remaining: 2.91s
489:    learn: 0.4441452        total: 2.79s    remaining: 2.91s
490:    learn: 0.4439810        total: 2.8s     remaining: 2.9s
491:    learn: 0.4435880        total: 2.81s    remaining: 2.9s
492:    learn: 0.4434965        total: 2.81s    remaining: 2.89s
493:    learn: 0.4432718        total: 2.82s    remaining: 2.88s
494:    learn: 0.4431939        total: 2.82s    remaining: 2.88s
495:    learn: 0.4430198        total: 2.83s    remaining: 2.87s
496:    learn: 0.4429146        total: 2.83s    remaining: 2.87s
497:    learn: 0.4426842        total: 2.84s    remaining: 2.86s
498:    learn: 0.4424525        total: 2.84s    remaining: 2.85s
499:    learn: 0.4422173        total: 2.85s    remaining: 2.85s
500:    learn: 0.4420523        total: 2.85s    remaining: 2.84s
501:    learn: 0.4417395        total: 2.86s    remaining: 2.83s
502:    learn: 0.4415963        total: 2.86s    remaining: 2.83s
503:    learn: 0.4414099        total: 2.87s    remaining: 2.82s
504:    learn: 0.4410848        total: 2.87s    remaining: 2.82s
505:    learn: 0.4409293        total: 2.88s    remaining: 2.81s
506:    learn: 0.4405828        total: 2.88s    remaining: 2.81s
507:    learn: 0.4404194        total: 2.89s    remaining: 2.8s
508:    learn: 0.4402853        total: 2.9s     remaining: 2.79s
509:    learn: 0.4401270        total: 2.9s     remaining: 2.79s
510:    learn: 0.4399366        total: 2.91s    remaining: 2.78s
511:    learn: 0.4397143        total: 2.91s    remaining: 2.77s
512:    learn: 0.4394688        total: 2.92s    remaining: 2.77s
513:    learn: 0.4393859        total: 2.92s    remaining: 2.76s
514:    learn: 0.4391658        total: 2.93s    remaining: 2.76s
515:    learn: 0.4386740        total: 2.93s    remaining: 2.75s
516:    learn: 0.4386633        total: 2.94s    remaining: 2.74s
517:    learn: 0.4384848        total: 2.94s    remaining: 2.74s
518:    learn: 0.4381980        total: 2.95s    remaining: 2.73s
519:    learn: 0.4377592        total: 2.95s    remaining: 2.73s
520:    learn: 0.4377022        total: 2.96s    remaining: 2.72s
521:    learn: 0.4375411        total: 2.96s    remaining: 2.71s
522:    learn: 0.4371467        total: 2.97s    remaining: 2.71s
523:    learn: 0.4370132        total: 2.98s    remaining: 2.7s
524:    learn: 0.4366771        total: 2.98s    remaining: 2.7s
525:    learn: 0.4364016        total: 2.98s    remaining: 2.69s
526:    learn: 0.4359668        total: 2.99s    remaining: 2.68s
527:    learn: 0.4358387        total: 3s       remaining: 2.68s
528:    learn: 0.4356973        total: 3s       remaining: 2.67s
529:    learn: 0.4353904        total: 3.01s    remaining: 2.67s
530:    learn: 0.4350741        total: 3.01s    remaining: 2.66s
531:    learn: 0.4348213        total: 3.02s    remaining: 2.65s
532:    learn: 0.4345818        total: 3.02s    remaining: 2.65s
533:    learn: 0.4343929        total: 3.03s    remaining: 2.64s
534:    learn: 0.4342821        total: 3.03s    remaining: 2.64s
535:    learn: 0.4340483        total: 3.04s    remaining: 2.63s
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536:    learn: 0.4337605        total: 3.04s    remaining: 2.63s
537:    learn: 0.4337443        total: 3.05s    remaining: 2.62s
538:    learn: 0.4336586        total: 3.06s    remaining: 2.61s
539:    learn: 0.4335776        total: 3.06s    remaining: 2.61s
540:    learn: 0.4335706        total: 3.06s    remaining: 2.6s
541:    learn: 0.4333774        total: 3.07s    remaining: 2.59s
542:    learn: 0.4332381        total: 3.08s    remaining: 2.59s
543:    learn: 0.4330632        total: 3.08s    remaining: 2.58s
544:    learn: 0.4327191        total: 3.09s    remaining: 2.58s
545:    learn: 0.4324112        total: 3.09s    remaining: 2.57s
546:    learn: 0.4323789        total: 3.1s     remaining: 2.56s
547:    learn: 0.4322521        total: 3.1s     remaining: 2.56s
548:    learn: 0.4321231        total: 3.11s    remaining: 2.55s
549:    learn: 0.4319795        total: 3.11s    remaining: 2.55s
550:    learn: 0.4316681        total: 3.12s    remaining: 2.54s
551:    learn: 0.4315494        total: 3.12s    remaining: 2.53s
552:    learn: 0.4314018        total: 3.13s    remaining: 2.53s
553:    learn: 0.4313054        total: 3.13s    remaining: 2.52s
554:    learn: 0.4311681        total: 3.14s    remaining: 2.52s
555:    learn: 0.4308395        total: 3.14s    remaining: 2.51s
556:    learn: 0.4306826        total: 3.15s    remaining: 2.5s
557:    learn: 0.4304480        total: 3.15s    remaining: 2.5s
558:    learn: 0.4301717        total: 3.16s    remaining: 2.49s
559:    learn: 0.4300155        total: 3.17s    remaining: 2.49s
560:    learn: 0.4297677        total: 3.17s    remaining: 2.48s
561:    learn: 0.4295124        total: 3.18s    remaining: 2.48s
562:    learn: 0.4292324        total: 3.18s    remaining: 2.47s
563:    learn: 0.4288152        total: 3.19s    remaining: 2.46s
564:    learn: 0.4286475        total: 3.19s    remaining: 2.46s
565:    learn: 0.4285487        total: 3.2s     remaining: 2.45s
566:    learn: 0.4283164        total: 3.2s     remaining: 2.45s
567:    learn: 0.4279931        total: 3.21s    remaining: 2.44s
568:    learn: 0.4279658        total: 3.21s    remaining: 2.44s
569:    learn: 0.4277445        total: 3.22s    remaining: 2.43s
570:    learn: 0.4275801        total: 3.23s    remaining: 2.42s
571:    learn: 0.4275320        total: 3.23s    remaining: 2.42s
572:    learn: 0.4274141        total: 3.24s    remaining: 2.42s
573:    learn: 0.4273264        total: 3.25s    remaining: 2.41s
574:    learn: 0.4272121        total: 3.25s    remaining: 2.4s
575:    learn: 0.4269876        total: 3.26s    remaining: 2.4s
576:    learn: 0.4266975        total: 3.27s    remaining: 2.39s
577:    learn: 0.4264419        total: 3.27s    remaining: 2.39s
578:    learn: 0.4260900        total: 3.28s    remaining: 2.38s
579:    learn: 0.4260365        total: 3.29s    remaining: 2.38s
580:    learn: 0.4259602        total: 3.29s    remaining: 2.37s
581:    learn: 0.4259537        total: 3.3s     remaining: 2.37s
582:    learn: 0.4257600        total: 3.31s    remaining: 2.37s
583:    learn: 0.4255662        total: 3.31s    remaining: 2.36s
584:    learn: 0.4254182        total: 3.32s    remaining: 2.35s
585:    learn: 0.4250862        total: 3.33s    remaining: 2.35s
586:    learn: 0.4250661        total: 3.33s    remaining: 2.34s
587:    learn: 0.4248790        total: 3.34s    remaining: 2.34s
588:    learn: 0.4247546        total: 3.34s    remaining: 2.33s
589:    learn: 0.4245846        total: 3.35s    remaining: 2.33s
590:    learn: 0.4245614        total: 3.35s    remaining: 2.32s
591:    learn: 0.4244366        total: 3.36s    remaining: 2.32s
592:    learn: 0.4242676        total: 3.37s    remaining: 2.31s
593:    learn: 0.4241234        total: 3.37s    remaining: 2.31s
594:    learn: 0.4240143        total: 3.38s    remaining: 2.3s
595:    learn: 0.4237697        total: 3.38s    remaining: 2.29s
596:    learn: 0.4236304        total: 3.39s    remaining: 2.29s
597:    learn: 0.4234844        total: 3.39s    remaining: 2.28s
598:    learn: 0.4234256        total: 3.4s     remaining: 2.28s
599:    learn: 0.4232476        total: 3.4s     remaining: 2.27s
600:    learn: 0.4230453        total: 3.41s    remaining: 2.26s
601:    learn: 0.4228436        total: 3.42s    remaining: 2.26s
602:    learn: 0.4226856        total: 3.42s    remaining: 2.25s
603:    learn: 0.4224797        total: 3.43s    remaining: 2.25s
604:    learn: 0.4220851        total: 3.43s    remaining: 2.24s
605:    learn: 0.4218299        total: 3.44s    remaining: 2.23s
606:    learn: 0.4216893        total: 3.44s    remaining: 2.23s
607:    learn: 0.4215178        total: 3.45s    remaining: 2.23s
608:    learn: 0.4212442        total: 3.46s    remaining: 2.22s
609:    learn: 0.4211081        total: 3.47s    remaining: 2.22s
610:    learn: 0.4209898        total: 3.47s    remaining: 2.21s
611:    learn: 0.4208935        total: 3.48s    remaining: 2.21s
612:    learn: 0.4206684        total: 3.49s    remaining: 2.2s
613:    learn: 0.4206496        total: 3.49s    remaining: 2.2s
614:    learn: 0.4206451        total: 3.5s     remaining: 2.19s
615:    learn: 0.4204357        total: 3.5s     remaining: 2.18s
616:    learn: 0.4203051        total: 3.51s    remaining: 2.18s
617:    learn: 0.4201521        total: 3.52s    remaining: 2.17s
618:    learn: 0.4198861        total: 3.52s    remaining: 2.17s
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619:    learn: 0.4195495        total: 3.52s    remaining: 2.16s
620:    learn: 0.4194913        total: 3.53s    remaining: 2.15s
621:    learn: 0.4193218        total: 3.54s    remaining: 2.15s
622:    learn: 0.4190737        total: 3.54s    remaining: 2.14s
623:    learn: 0.4188798        total: 3.55s    remaining: 2.14s
624:    learn: 0.4185356        total: 3.55s    remaining: 2.13s
625:    learn: 0.4184447        total: 3.56s    remaining: 2.13s
626:    learn: 0.4181815        total: 3.56s    remaining: 2.12s
627:    learn: 0.4180906        total: 3.57s    remaining: 2.11s
628:    learn: 0.4179299        total: 3.57s    remaining: 2.11s
629:    learn: 0.4177341        total: 3.58s    remaining: 2.1s
630:    learn: 0.4175470        total: 3.58s    remaining: 2.1s
631:    learn: 0.4173829        total: 3.59s    remaining: 2.09s
632:    learn: 0.4172254        total: 3.59s    remaining: 2.08s
633:    learn: 0.4169947        total: 3.6s     remaining: 2.08s
634:    learn: 0.4167829        total: 3.6s     remaining: 2.07s
635:    learn: 0.4165655        total: 3.61s    remaining: 2.07s
636:    learn: 0.4163934        total: 3.62s    remaining: 2.06s
637:    learn: 0.4163415        total: 3.62s    remaining: 2.05s
638:    learn: 0.4159349        total: 3.63s    remaining: 2.05s
639:    learn: 0.4157224        total: 3.63s    remaining: 2.04s
640:    learn: 0.4156320        total: 3.63s    remaining: 2.04s
641:    learn: 0.4153324        total: 3.64s    remaining: 2.03s
642:    learn: 0.4150268        total: 3.65s    remaining: 2.02s
643:    learn: 0.4148364        total: 3.65s    remaining: 2.02s
644:    learn: 0.4146060        total: 3.66s    remaining: 2.01s
645:    learn: 0.4145268        total: 3.66s    remaining: 2.01s
646:    learn: 0.4142022        total: 3.67s    remaining: 2s
647:    learn: 0.4140647        total: 3.67s    remaining: 2s
648:    learn: 0.4139137        total: 3.68s    remaining: 1.99s
649:    learn: 0.4135296        total: 3.68s    remaining: 1.98s
650:    learn: 0.4134264        total: 3.69s    remaining: 1.98s
651:    learn: 0.4131412        total: 3.69s    remaining: 1.97s
652:    learn: 0.4129923        total: 3.7s     remaining: 1.97s
653:    learn: 0.4129607        total: 3.7s     remaining: 1.96s
654:    learn: 0.4128248        total: 3.71s    remaining: 1.95s
655:    learn: 0.4127432        total: 3.71s    remaining: 1.95s
656:    learn: 0.4124864        total: 3.72s    remaining: 1.94s
657:    learn: 0.4122231        total: 3.73s    remaining: 1.94s
658:    learn: 0.4119524        total: 3.73s    remaining: 1.93s
659:    learn: 0.4117798        total: 3.74s    remaining: 1.93s
660:    learn: 0.4116387        total: 3.74s    remaining: 1.92s
661:    learn: 0.4114917        total: 3.75s    remaining: 1.91s
662:    learn: 0.4112515        total: 3.75s    remaining: 1.91s
663:    learn: 0.4111863        total: 3.76s    remaining: 1.9s
664:    learn: 0.4109868        total: 3.76s    remaining: 1.9s
665:    learn: 0.4107792        total: 3.77s    remaining: 1.89s
666:    learn: 0.4107747        total: 3.77s    remaining: 1.88s
667:    learn: 0.4106419        total: 3.78s    remaining: 1.88s
668:    learn: 0.4105077        total: 3.79s    remaining: 1.87s
669:    learn: 0.4105025        total: 3.79s    remaining: 1.87s
670:    learn: 0.4104059        total: 3.8s     remaining: 1.86s
671:    learn: 0.4102343        total: 3.8s     remaining: 1.85s
672:    learn: 0.4100719        total: 3.81s    remaining: 1.85s
673:    learn: 0.4097504        total: 3.81s    remaining: 1.84s
674:    learn: 0.4095387        total: 3.82s    remaining: 1.84s
675:    learn: 0.4092908        total: 3.82s    remaining: 1.83s
676:    learn: 0.4090860        total: 3.83s    remaining: 1.83s
677:    learn: 0.4089029        total: 3.83s    remaining: 1.82s
678:    learn: 0.4086640        total: 3.84s    remaining: 1.81s
679:    learn: 0.4085241        total: 3.84s    remaining: 1.81s
680:    learn: 0.4084366        total: 3.85s    remaining: 1.8s
681:    learn: 0.4081647        total: 3.85s    remaining: 1.8s
682:    learn: 0.4079980        total: 3.86s    remaining: 1.79s
683:    learn: 0.4077907        total: 3.87s    remaining: 1.78s
684:    learn: 0.4077787        total: 3.87s    remaining: 1.78s
685:    learn: 0.4075996        total: 3.88s    remaining: 1.77s
686:    learn: 0.4073477        total: 3.88s    remaining: 1.77s
687:    learn: 0.4072866        total: 3.89s    remaining: 1.76s
688:    learn: 0.4070165        total: 3.89s    remaining: 1.76s
689:    learn: 0.4067922        total: 3.9s     remaining: 1.75s
690:    learn: 0.4067856        total: 3.9s     remaining: 1.75s
691:    learn: 0.4067815        total: 3.91s    remaining: 1.74s
692:    learn: 0.4065480        total: 3.91s    remaining: 1.73s
693:    learn: 0.4064209        total: 3.92s    remaining: 1.73s
694:    learn: 0.4064173        total: 3.92s    remaining: 1.72s
695:    learn: 0.4063868        total: 3.93s    remaining: 1.72s
696:    learn: 0.4062482        total: 3.94s    remaining: 1.71s
697:    learn: 0.4060159        total: 3.94s    remaining: 1.71s
698:    learn: 0.4059906        total: 3.95s    remaining: 1.7s
699:    learn: 0.4059352        total: 3.95s    remaining: 1.69s
700:    learn: 0.4056344        total: 3.96s    remaining: 1.69s
701:    learn: 0.4054971        total: 3.96s    remaining: 1.68s
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702:    learn: 0.4053213        total: 3.97s    remaining: 1.68s
703:    learn: 0.4051203        total: 3.97s    remaining: 1.67s
704:    learn: 0.4049821        total: 3.98s    remaining: 1.66s
705:    learn: 0.4048665        total: 3.98s    remaining: 1.66s
706:    learn: 0.4047100        total: 3.99s    remaining: 1.65s
707:    learn: 0.4046246        total: 3.99s    remaining: 1.65s
708:    learn: 0.4044857        total: 4s       remaining: 1.64s
709:    learn: 0.4042322        total: 4s       remaining: 1.64s
710:    learn: 0.4039043        total: 4.01s    remaining: 1.63s
711:    learn: 0.4038687        total: 4.01s    remaining: 1.62s
712:    learn: 0.4035484        total: 4.02s    remaining: 1.62s
713:    learn: 0.4035463        total: 4.03s    remaining: 1.61s
714:    learn: 0.4033430        total: 4.03s    remaining: 1.61s
715:    learn: 0.4032411        total: 4.04s    remaining: 1.6s
716:    learn: 0.4030668        total: 4.04s    remaining: 1.59s
717:    learn: 0.4029219        total: 4.05s    remaining: 1.59s
718:    learn: 0.4026446        total: 4.05s    remaining: 1.58s
719:    learn: 0.4023357        total: 4.06s    remaining: 1.58s
720:    learn: 0.4020822        total: 4.06s    remaining: 1.57s
721:    learn: 0.4019615        total: 4.07s    remaining: 1.57s
722:    learn: 0.4018499        total: 4.07s    remaining: 1.56s
723:    learn: 0.4016635        total: 4.08s    remaining: 1.55s
724:    learn: 0.4015354        total: 4.08s    remaining: 1.55s
725:    learn: 0.4014333        total: 4.09s    remaining: 1.54s
726:    learn: 0.4012230        total: 4.09s    remaining: 1.54s
727:    learn: 0.4011085        total: 4.1s     remaining: 1.53s
728:    learn: 0.4008581        total: 4.11s    remaining: 1.53s
729:    learn: 0.4007269        total: 4.11s    remaining: 1.52s
730:    learn: 0.4004890        total: 4.12s    remaining: 1.51s
731:    learn: 0.4003481        total: 4.12s    remaining: 1.51s
732:    learn: 0.4000871        total: 4.13s    remaining: 1.5s
733:    learn: 0.3998218        total: 4.13s    remaining: 1.5s
734:    learn: 0.3995548        total: 4.14s    remaining: 1.49s
735:    learn: 0.3994042        total: 4.15s    remaining: 1.49s
736:    learn: 0.3991288        total: 4.15s    remaining: 1.48s
737:    learn: 0.3988358        total: 4.16s    remaining: 1.48s
738:    learn: 0.3987438        total: 4.16s    remaining: 1.47s
739:    learn: 0.3984609        total: 4.17s    remaining: 1.47s
740:    learn: 0.3983518        total: 4.18s    remaining: 1.46s
741:    learn: 0.3983380        total: 4.18s    remaining: 1.45s
742:    learn: 0.3981534        total: 4.19s    remaining: 1.45s
743:    learn: 0.3981189        total: 4.19s    remaining: 1.44s
744:    learn: 0.3979712        total: 4.2s     remaining: 1.44s
745:    learn: 0.3976881        total: 4.2s     remaining: 1.43s
746:    learn: 0.3974513        total: 4.21s    remaining: 1.43s
747:    learn: 0.3973485        total: 4.21s    remaining: 1.42s
748:    learn: 0.3972199        total: 4.22s    remaining: 1.41s
749:    learn: 0.3970623        total: 4.23s    remaining: 1.41s
750:    learn: 0.3969509        total: 4.23s    remaining: 1.4s
751:    learn: 0.3968987        total: 4.24s    remaining: 1.4s
752:    learn: 0.3966524        total: 4.25s    remaining: 1.39s
753:    learn: 0.3965501        total: 4.25s    remaining: 1.39s
754:    learn: 0.3963028        total: 4.26s    remaining: 1.38s
755:    learn: 0.3961565        total: 4.26s    remaining: 1.38s
756:    learn: 0.3960950        total: 4.27s    remaining: 1.37s
757:    learn: 0.3960917        total: 4.27s    remaining: 1.36s
758:    learn: 0.3958547        total: 4.28s    remaining: 1.36s
759:    learn: 0.3958506        total: 4.29s    remaining: 1.35s
760:    learn: 0.3958469        total: 4.29s    remaining: 1.35s
761:    learn: 0.3956386        total: 4.29s    remaining: 1.34s
762:    learn: 0.3953488        total: 4.3s     remaining: 1.33s
763:    learn: 0.3951204        total: 4.31s    remaining: 1.33s
764:    learn: 0.3949599        total: 4.31s    remaining: 1.32s
765:    learn: 0.3947599        total: 4.32s    remaining: 1.32s
766:    learn: 0.3947363        total: 4.32s    remaining: 1.31s
767:    learn: 0.3945722        total: 4.33s    remaining: 1.31s
768:    learn: 0.3944468        total: 4.33s    remaining: 1.3s
769:    learn: 0.3942194        total: 4.34s    remaining: 1.29s
770:    learn: 0.3940593        total: 4.34s    remaining: 1.29s
771:    learn: 0.3939863        total: 4.35s    remaining: 1.28s
772:    learn: 0.3938289        total: 4.35s    remaining: 1.28s
773:    learn: 0.3935558        total: 4.36s    remaining: 1.27s
774:    learn: 0.3932999        total: 4.36s    remaining: 1.27s
775:    learn: 0.3931600        total: 4.37s    remaining: 1.26s
776:    learn: 0.3930678        total: 4.38s    remaining: 1.25s
777:    learn: 0.3930629        total: 4.38s    remaining: 1.25s
778:    learn: 0.3929446        total: 4.38s    remaining: 1.24s
779:    learn: 0.3926652        total: 4.39s    remaining: 1.24s
780:    learn: 0.3924967        total: 4.39s    remaining: 1.23s
781:    learn: 0.3923536        total: 4.4s     remaining: 1.23s
782:    learn: 0.3920846        total: 4.41s    remaining: 1.22s
783:    learn: 0.3920814        total: 4.41s    remaining: 1.22s
784:    learn: 0.3918177        total: 4.42s    remaining: 1.21s
```

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785:    learn: 0.3915791        total: 4.42s    remaining: 1.2s
786:    learn: 0.3914714        total: 4.43s    remaining: 1.2s
787:    learn: 0.3913901        total: 4.43s    remaining: 1.19s
788:    learn: 0.3912828        total: 4.44s    remaining: 1.19s
789:    learn: 0.3911974        total: 4.44s    remaining: 1.18s
790:    learn: 0.3910324        total: 4.45s    remaining: 1.18s
791:    learn: 0.3909211        total: 4.45s    remaining: 1.17s
792:    learn: 0.3907448        total: 4.46s    remaining: 1.16s
793:    learn: 0.3904480        total: 4.46s    remaining: 1.16s
794:    learn: 0.3902781        total: 4.47s    remaining: 1.15s
795:    learn: 0.3901227        total: 4.48s    remaining: 1.15s
796:    learn: 0.3898630        total: 4.48s    remaining: 1.14s
797:    learn: 0.3898027        total: 4.49s    remaining: 1.14s
798:    learn: 0.3897082        total: 4.49s    remaining: 1.13s
799:    learn: 0.3895103        total: 4.5s     remaining: 1.12s
800:    learn: 0.3894741        total: 4.5s     remaining: 1.12s
801:    learn: 0.3893448        total: 4.51s    remaining: 1.11s
802:    learn: 0.3893131        total: 4.51s    remaining: 1.11s
803:    learn: 0.3890194        total: 4.52s    remaining: 1.1s
804:    learn: 0.3887978        total: 4.53s    remaining: 1.1s
805:    learn: 0.3887137        total: 4.53s    remaining: 1.09s
806:    learn: 0.3886545        total: 4.54s    remaining: 1.08s
807:    learn: 0.3886523        total: 4.54s    remaining: 1.08s
808:    learn: 0.3885767        total: 4.55s    remaining: 1.07s
809:    learn: 0.3884901        total: 4.55s    remaining: 1.07s
810:    learn: 0.3883600        total: 4.56s    remaining: 1.06s
811:    learn: 0.3882939        total: 4.56s    remaining: 1.06s
812:    learn: 0.3881404        total: 4.57s    remaining: 1.05s
813:    learn: 0.3878887        total: 4.57s    remaining: 1.04s
814:    learn: 0.3878168        total: 4.58s    remaining: 1.04s
815:    learn: 0.3877461        total: 4.58s    remaining: 1.03s
816:    learn: 0.3876855        total: 4.59s    remaining: 1.03s
817:    learn: 0.3875705        total: 4.59s    remaining: 1.02s
818:    learn: 0.3874676        total: 4.6s     remaining: 1.02s
819:    learn: 0.3874566        total: 4.6s     remaining: 1.01s
820:    learn: 0.3874009        total: 4.61s    remaining: 1s
821:    learn: 0.3873394        total: 4.61s    remaining: 999ms
822:    learn: 0.3873140        total: 4.62s    remaining: 993ms
823:    learn: 0.3873114        total: 4.62s    remaining: 988ms
824:    learn: 0.3871360        total: 4.63s    remaining: 982ms
825:    learn: 0.3871014        total: 4.63s    remaining: 976ms
826:    learn: 0.3869637        total: 4.64s    remaining: 971ms
827:    learn: 0.3869411        total: 4.64s    remaining: 965ms
828:    learn: 0.3866656        total: 4.65s    remaining: 960ms
829:    learn: 0.3866635        total: 4.66s    remaining: 954ms
830:    learn: 0.3864559        total: 4.66s    remaining: 948ms
831:    learn: 0.3861926        total: 4.67s    remaining: 943ms
832:    learn: 0.3858617        total: 4.67s    remaining: 937ms
833:    learn: 0.3857645        total: 4.68s    remaining: 931ms
834:    learn: 0.3856934        total: 4.68s    remaining: 925ms
835:    learn: 0.3855704        total: 4.69s    remaining: 920ms
836:    learn: 0.3854310        total: 4.69s    remaining: 914ms
837:    learn: 0.3852302        total: 4.7s     remaining: 909ms
838:    learn: 0.3850761        total: 4.71s    remaining: 903ms
839:    learn: 0.3849434        total: 4.71s    remaining: 897ms
840:    learn: 0.3846675        total: 4.72s    remaining: 892ms
841:    learn: 0.3845414        total: 4.72s    remaining: 886ms
842:    learn: 0.3842390        total: 4.73s    remaining: 880ms
843:    learn: 0.3841424        total: 4.73s    remaining: 875ms
844:    learn: 0.3840415        total: 4.74s    remaining: 869ms
845:    learn: 0.3838276        total: 4.74s    remaining: 863ms
846:    learn: 0.3836565        total: 4.75s    remaining: 858ms
847:    learn: 0.3834964        total: 4.75s    remaining: 852ms
848:    learn: 0.3833760        total: 4.76s    remaining: 846ms
849:    learn: 0.3832683        total: 4.76s    remaining: 841ms
850:    learn: 0.3831477        total: 4.77s    remaining: 835ms
851:    learn: 0.3830318        total: 4.77s    remaining: 829ms
852:    learn: 0.3829993        total: 4.78s    remaining: 824ms
853:    learn: 0.3828521        total: 4.79s    remaining: 818ms
854:    learn: 0.3827148        total: 4.79s    remaining: 812ms
855:    learn: 0.3825811        total: 4.79s    remaining: 807ms
856:    learn: 0.3825064        total: 4.8s     remaining: 801ms
857:    learn: 0.3823917        total: 4.81s    remaining: 795ms
858:    learn: 0.3820498        total: 4.81s    remaining: 790ms
859:    learn: 0.3819676        total: 4.82s    remaining: 784ms
860:    learn: 0.3818354        total: 4.82s    remaining: 778ms
861:    learn: 0.3816178        total: 4.83s    remaining: 773ms
862:    learn: 0.3813418        total: 4.83s    remaining: 767ms
863:    learn: 0.3811299        total: 4.84s    remaining: 761ms
864:    learn: 0.3810420        total: 4.84s    remaining: 756ms
865:    learn: 0.3809617        total: 4.85s    remaining: 750ms
866:    learn: 0.3807917        total: 4.85s    remaining: 745ms
867:    learn: 0.3806194        total: 4.86s    remaining: 739ms
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868:    learn: 0.3804815        total: 4.86s    remaining: 733ms
869:    learn: 0.3802715        total: 4.87s    remaining: 728ms
870:    learn: 0.3801790        total: 4.87s    remaining: 722ms
871:    learn: 0.3801771        total: 4.88s    remaining: 716ms
872:    learn: 0.3801217        total: 4.88s    remaining: 711ms
873:    learn: 0.3798206        total: 4.89s    remaining: 705ms
874:    learn: 0.3797699        total: 4.89s    remaining: 699ms
875:    learn: 0.3796504        total: 4.9s     remaining: 694ms
876:    learn: 0.3794435        total: 4.91s    remaining: 688ms
877:    learn: 0.3792172        total: 4.91s    remaining: 682ms
878:    learn: 0.3790870        total: 4.92s    remaining: 677ms
879:    learn: 0.3788802        total: 4.92s    remaining: 671ms
880:    learn: 0.3787253        total: 4.93s    remaining: 666ms
881:    learn: 0.3786014        total: 4.93s    remaining: 660ms
882:    learn: 0.3783627        total: 4.94s    remaining: 655ms
883:    learn: 0.3782741        total: 4.95s    remaining: 649ms
884:    learn: 0.3781935        total: 4.95s    remaining: 643ms
885:    learn: 0.3780043        total: 4.96s    remaining: 638ms
886:    learn: 0.3779174        total: 4.96s    remaining: 632ms
887:    learn: 0.3778581        total: 4.97s    remaining: 626ms
888:    learn: 0.3776365        total: 4.97s    remaining: 621ms
889:    learn: 0.3775402        total: 4.98s    remaining: 615ms
890:    learn: 0.3774753        total: 4.98s    remaining: 610ms
891:    learn: 0.3773226        total: 4.99s    remaining: 604ms
892:    learn: 0.3771829        total: 4.99s    remaining: 598ms
893:    learn: 0.3770049        total: 5s       remaining: 593ms
894:    learn: 0.3769109        total: 5s       remaining: 587ms
895:    learn: 0.3766721        total: 5.01s    remaining: 582ms
896:    learn: 0.3765630        total: 5.01s    remaining: 576ms
897:    learn: 0.3764319        total: 5.02s    remaining: 570ms
898:    learn: 0.3762577        total: 5.03s    remaining: 565ms
899:    learn: 0.3761863        total: 5.03s    remaining: 559ms
900:    learn: 0.3760639        total: 5.04s    remaining: 553ms
901:    learn: 0.3758682        total: 5.04s    remaining: 548ms
902:    learn: 0.3756938        total: 5.05s    remaining: 542ms
903:    learn: 0.3755484        total: 5.05s    remaining: 537ms
904:    learn: 0.3754402        total: 5.06s    remaining: 531ms
905:    learn: 0.3753049        total: 5.06s    remaining: 525ms
906:    learn: 0.3752745        total: 5.07s    remaining: 520ms
907:    learn: 0.3751418        total: 5.07s    remaining: 514ms
908:    learn: 0.3749874        total: 5.08s    remaining: 508ms
909:    learn: 0.3747856        total: 5.08s    remaining: 503ms
910:    learn: 0.3746933        total: 5.09s    remaining: 497ms
911:    learn: 0.3745674        total: 5.09s    remaining: 492ms
912:    learn: 0.3744953        total: 5.1s     remaining: 486ms
913:    learn: 0.3742943        total: 5.1s     remaining: 480ms
914:    learn: 0.3741117        total: 5.11s    remaining: 475ms
915:    learn: 0.3741050        total: 5.12s    remaining: 469ms
916:    learn: 0.3740230        total: 5.12s    remaining: 464ms
917:    learn: 0.3739363        total: 5.13s    remaining: 458ms
918:    learn: 0.3738849        total: 5.13s    remaining: 452ms
919:    learn: 0.3738263        total: 5.14s    remaining: 447ms
920:    learn: 0.3735970        total: 5.14s    remaining: 441ms
921:    learn: 0.3735440        total: 5.15s    remaining: 436ms
922:    learn: 0.3735234        total: 5.16s    remaining: 430ms
923:    learn: 0.3732307        total: 5.16s    remaining: 424ms
924:    learn: 0.3730927        total: 5.17s    remaining: 419ms
925:    learn: 0.3730340        total: 5.17s    remaining: 413ms
926:    learn: 0.3728714        total: 5.18s    remaining: 408ms
927:    learn: 0.3727398        total: 5.18s    remaining: 402ms
928:    learn: 0.3726155        total: 5.19s    remaining: 396ms
929:    learn: 0.3725001        total: 5.19s    remaining: 391ms
930:    learn: 0.3724328        total: 5.2s     remaining: 385ms
931:    learn: 0.3723378        total: 5.2s     remaining: 380ms
932:    learn: 0.3721244        total: 5.21s    remaining: 374ms
933:    learn: 0.3718663        total: 5.21s    remaining: 368ms
934:    learn: 0.3717362        total: 5.22s    remaining: 363ms
935:    learn: 0.3717196        total: 5.23s    remaining: 357ms
936:    learn: 0.3715457        total: 5.23s    remaining: 352ms
937:    learn: 0.3713033        total: 5.24s    remaining: 346ms
938:    learn: 0.3711694        total: 5.24s    remaining: 341ms
939:    learn: 0.3710517        total: 5.25s    remaining: 335ms
940:    learn: 0.3710017        total: 5.25s    remaining: 330ms
941:    learn: 0.3707743        total: 5.26s    remaining: 324ms
942:    learn: 0.3706061        total: 5.27s    remaining: 318ms
943:    learn: 0.3705301        total: 5.27s    remaining: 313ms
944:    learn: 0.3704764        total: 5.28s    remaining: 307ms
945:    learn: 0.3702647        total: 5.29s    remaining: 302ms
946:    learn: 0.3700572        total: 5.29s    remaining: 296ms
947:    learn: 0.3698871        total: 5.3s     remaining: 291ms
948:    learn: 0.3697242        total: 5.31s    remaining: 285ms
949:    learn: 0.3696946        total: 5.31s    remaining: 280ms
950:    learn: 0.3695365        total: 5.32s    remaining: 274ms
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951:    learn: 0.3694323        total: 5.33s    remaining: 269ms
952:    learn: 0.3691789        total: 5.33s    remaining: 263ms
953:    learn: 0.3690255        total: 5.34s    remaining: 257ms
954:    learn: 0.3688992        total: 5.35s    remaining: 252ms
955:    learn: 0.3687226        total: 5.36s    remaining: 246ms
956:    learn: 0.3686238        total: 5.37s    remaining: 241ms
957:    learn: 0.3683476        total: 5.38s    remaining: 236ms
958:    learn: 0.3681776        total: 5.38s    remaining: 230ms
959:    learn: 0.3681245        total: 5.39s    remaining: 225ms
960:    learn: 0.3679949        total: 5.4s     remaining: 219ms
961:    learn: 0.3677986        total: 5.4s     remaining: 213ms
962:    learn: 0.3676300        total: 5.41s    remaining: 208ms
963:    learn: 0.3675226        total: 5.41s    remaining: 202ms
964:    learn: 0.3673618        total: 5.42s    remaining: 197ms
965:    learn: 0.3672139        total: 5.42s    remaining: 191ms
966:    learn: 0.3671139        total: 5.43s    remaining: 185ms
967:    learn: 0.3669944        total: 5.43s    remaining: 180ms
968:    learn: 0.3668290        total: 5.44s    remaining: 174ms
969:    learn: 0.3666016        total: 5.45s    remaining: 168ms
970:    learn: 0.3663678        total: 5.45s    remaining: 163ms
971:    learn: 0.3662295        total: 5.46s    remaining: 157ms
972:    learn: 0.3660710        total: 5.46s    remaining: 152ms
973:    learn: 0.3660302        total: 5.47s    remaining: 146ms
974:    learn: 0.3659869        total: 5.47s    remaining: 140ms
975:    learn: 0.3658439        total: 5.48s    remaining: 135ms
976:    learn: 0.3656260        total: 5.48s    remaining: 129ms
977:    learn: 0.3654638        total: 5.49s    remaining: 123ms
978:    learn: 0.3653482        total: 5.49s    remaining: 118ms
979:    learn: 0.3653075        total: 5.5s     remaining: 112ms
980:    learn: 0.3651763        total: 5.5s     remaining: 107ms
981:    learn: 0.3650175        total: 5.51s    remaining: 101ms
982:    learn: 0.3648098        total: 5.51s    remaining: 95.4ms
983:    learn: 0.3646888        total: 5.52s    remaining: 89.8ms
984:    learn: 0.3645928        total: 5.53s    remaining: 84.2ms
985:    learn: 0.3644555        total: 5.53s    remaining: 78.6ms
986:    learn: 0.3643043        total: 5.54s    remaining: 72.9ms
987:    learn: 0.3643033        total: 5.54s    remaining: 67.3ms
988:    learn: 0.3641526        total: 5.55s    remaining: 61.7ms
989:    learn: 0.3640723        total: 5.55s    remaining: 56.1ms
990:    learn: 0.3639925        total: 5.56s    remaining: 50.5ms
991:    learn: 0.3638391        total: 5.56s    remaining: 44.9ms
992:    learn: 0.3638378        total: 5.57s    remaining: 39.3ms
993:    learn: 0.3637319        total: 5.57s    remaining: 33.6ms
994:    learn: 0.3635421        total: 5.58s    remaining: 28ms
995:    learn: 0.3633855        total: 5.58s    remaining: 22.4ms
996:    learn: 0.3632862        total: 5.59s    remaining: 16.8ms
997:    learn: 0.3631167        total: 5.6s     remaining: 11.2ms
998:    learn: 0.3628914        total: 5.6s     remaining: 5.61ms
999:    learn: 0.3626985        total: 5.61s    remaining: 0us
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of testing was 0.000493 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 1638
[LightGBM] [Info] Number of data points in the train set: 4527, number of used features: 8
[LightGBM] [Info] Start training from score 5.901723
[iter 0] loss=1.7314 val_loss=0.0000 scale=1.0000 norm=1.2679
[iter 100] loss=1.2175 val_loss=0.0000 scale=2.0000 norm=1.5780
[iter 200] loss=0.8440 val_loss=0.0000 scale=2.0000 norm=1.3739
[iter 300] loss=0.7338 val_loss=0.0000 scale=1.0000 norm=0.6866
[iter 400] loss=0.6819 val_loss=0.0000 scale=1.0000 norm=0.6852
```

In [108…

```python
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score#

# Linear Regression
MSE_lr = mean_squared_error(y_test, y_pred_lr)
MAE_lr = mean_absolute_error(y_test, y_pred_lr)
R2_Score_lr = r2_score(y_test, y_pred_lr)
MAPE_lr = MAE_lr / np.mean(y_test)  # MAPE formula
Accuracy_lr = (1 - MAPE_lr) * 100  # Convert to percentage

print('   Performance Evaluation for Linear Regression Model: ')
print('Mean Squared Error (MSE): ', MSE_lr)
print('Mean Absolute Error (MAE): ', MAE_lr)
print('R² Score: ', R2_Score_lr)
print(f'Accuracy Percentage: {Accuracy_lr:.2f}%')

# Random Forest Model Evaluation
MSE_rf = mean_squared_error(y_test, y_pred_rf)
MAE_rf = mean_absolute_error(y_test, y_pred_rf)
R2_Score_rf = r2_score(y_test, y_pred_rf)
MAPE_rf = MAE_rf / np.mean(y_test)  # MAPE formula
Accuracy_rf = (1 - MAPE_rf) * 100  # Convert to percentage

print('\n   Performance Evaluation for Random Forest Model: ')
```

```
print('Mean Squared Error (MSE): ', MSE_rf)
print('Mean Absolute Error (MAE): ', MAE_rf)
print('R² Score: ', R2_Score_rf)
print(f'Accuracy Percentage: {Accuracy_rf:.2f}%')

# Gradient Boosting Model Evaluation
MSE_gb = mean_squared_error(y_test, y_pred_gb)
MAE_gb = mean_absolute_error(y_test, y_pred_gb)
R2_Score_gb = r2_score(y_test, y_pred_gb)
MAPE_gb = MAE_gb / np.mean(y_test)  # MAPE formula
Accuracy_gb = (1 - MAPE_gb) * 100  # Convert to percentage

print('\n   Performance Evaluation for Gradient Boosting Model: ')
print('Mean Squared Error (MSE): ', MSE_gb)
print('Mean Absolute Error (MAE): ', MAE_gb)
print('R² Score: ', R2_Score_gb)
print(f'Accuracy Percentage : {Accuracy_gb:.2f}%')

# XGBoost Model Evaluation
MSE_xgb = mean_squared_error(y_test, y_pred_xgb)
MAE_xgb = mean_absolute_error(y_test, y_pred_xgb)
R2_Score_xgb = r2_score(y_test, y_pred_xgb)
MAPE_xgb = MAE_xgb / np.mean(y_test)  # MAPE formula
Accuracy_xgb = (1 - MAPE_xgb) * 100  # Convert to percentage

print('\n   Performance Evaluation for XGBoost Model: ')
print('Mean Squared Error (MSE): ', MSE_xgb)
print('Mean Absolute Error (MAE): ', MAE_xgb)
print('R² Score: ', R2_Score_xgb)
print(f'Accuracy Percentage: {Accuracy_xgb:.2f}%')

# CatBoost Model Evaluation
MSE_cb = mean_squared_error(y_test, y_pred_cb)
MAE_cb = mean_absolute_error(y_test, y_pred_cb)
R2_Score_cb = r2_score(y_test, y_pred_cb)
MAPE_cb = MAE_cb / np.mean(y_test)  # MAPE formula
Accuracy_cb = (1 - MAPE_cb) * 100  # Convert to percentage

print('\n   Performance Evaluation for CatBoost Model: ')
print('Mean Squared Error (MSE): ', MSE_cb)
print('Mean Absolute Error (MAE): ', MAE_cb)
print('R² Score: ', R2_Score_cb)
print(f'Accuracy Percentage: {Accuracy_cb:.2f}%')

# AdaBoost Model Evaluation
MSE_ab = mean_squared_error(y_test, y_pred_ab)
MAE_ab = mean_absolute_error(y_test, y_pred_ab)
R2_Score_ab = r2_score(y_test, y_pred_ab)
MAPE_ab = MAE_ab / np.mean(y_test)  # MAPE formula
Accuracy_ab = (1 - MAPE_ab) * 100  # Convert to percentage

print('\n   Performance Evaluation for AdaBoost Model: ')
print('Mean Squared Error (MSE): ', MSE_ab)
print('Mean Absolute Error (MAE): ', MAE_ab)
print('R² Score: ', R2_Score_ab)
print(f'Accuracy Percentage: {Accuracy_ab:.2f}%')

# LightGBM Model Evaluation
MSE_lgb = mean_squared_error(y_test, y_pred_lgb)
MAE_lgb = mean_absolute_error(y_test, y_pred_lgb)
R2_Score_lgb = r2_score(y_test, y_pred_lgb)
MAPE_lgb = MAE_lgb / np.mean(y_test)  # MAPE formula
Accuracy_lgb = (1 - MAPE_lgb) * 100  # Convert to percentage

print('\n   Performance Evaluation for LightGBM Model: ')
print('Mean Squared Error (MSE): ', MSE_lgb)
print('Mean Absolute Error (MAE): ', MAE_lgb)
print('R² Score: ', R2_Score_lgb)
print(f'Accuracy Percentage: {Accuracy_lgb:.2f}%')

# NGBoost Model Evaluation
MSE_ngb = mean_squared_error(y_test, y_pred_ngb)
MAE_ngb = mean_absolute_error(y_test, y_pred_ngb)
R2_Score_ngb = r2_score(y_test, y_pred_ngb)
MAPE_ngb = MAE_ngb / np.mean(y_test)  # MAPE formula
Accuracy_ngb = (1 - MAPE_ngb) * 100  # Convert to percentage

print('\n   Performance Evaluation for NGBoost Model: ')
print('Mean Squared Error (MSE): ', MSE_ngb)
print('Mean Absolute Error (MAE): ', MAE_ngb)
print('R² Score: ', R2_Score_ngb)
print(f'Accuracy Percentage: {Accuracy_ngb:.2f}%')
```

```
# Stacking Regressor Model Evaluation
MSE_sr = mean_squared_error(y_test, y_pred_sr)
MAE_sr = mean_absolute_error(y_test, y_pred_sr)
R2_Score_sr = r2_score(y_test, y_pred_sr)
MAPE_sr = MAE_sr / np.mean(y_test)  # MAPE formula
Accuracy_sr = (1 - MAPE_sr) * 100  # Convert to percentage

print('\n   Performance Evaluation for Stacking Regressor Model: ')
print('Mean Squared Error (MSE): ', MSE_sr)
print('Mean Absolute Error (MAE): ', MAE_sr)
print('R² Score: ', R2_Score_sr)
print(f'Accuracy Percentage : {Accuracy_sr:.2f}%')
```

```
   Performance Evaluation for Linear Regression Model:
Mean Squared Error (MSE):  0.4128334767968585
Mean Absolute Error (MAE):  0.47485223510364033
R² Score:  0.7999866069727857
Accuracy Percentage: 91.93%

   Performance Evaluation for Random Forest Model:
Mean Squared Error (MSE):  0.31977202473498223
Mean Absolute Error (MAE):  0.38530388692579504
R² Score:  0.8450738826737686
Accuracy Percentage: 93.45%

   Performance Evaluation for Gradient Boosting Model:
Mean Squared Error (MSE):  0.31341593557267294
Mean Absolute Error (MAE):  0.41204892439883944
R² Score:  0.8481533397216827
Accuracy Percentage : 93.00%

   Performance Evaluation for XGBoost Model:
Mean Squared Error (MSE):  0.3242664273171274
Mean Absolute Error (MAE):  0.4001510744802522
R² Score:  0.8428963928125782
Accuracy Percentage: 93.20%

   Performance Evaluation for CatBoost Model:
Mean Squared Error (MSE):  0.2697698831985145
Mean Absolute Error (MAE):  0.3666807720664588
R² Score:  0.8692993841154968
Accuracy Percentage: 93.77%

   Performance Evaluation for AdaBoost Model:
Mean Squared Error (MSE):  0.5108575752553447
Mean Absolute Error (MAE):  0.5394307232799328
R² Score:  0.7524949823031053
Accuracy Percentage: 90.84%

   Performance Evaluation for LightGBM Model:
Mean Squared Error (MSE):  0.29025627516702107
Mean Absolute Error (MAE):  0.3800537049864135
R² Score:  0.859373946865836
Accuracy Percentage: 93.54%

   Performance Evaluation for NGBoost Model:
Mean Squared Error (MSE):  0.3302287286940039
Mean Absolute Error (MAE):  0.4198875616344445
R² Score:  0.840007721724437
Accuracy Percentage: 92.87%

   Performance Evaluation for Stacking Regressor Model:
Mean Squared Error (MSE):  0.2993638954112371
Mean Absolute Error (MAE):  0.3901567429572037
R² Score:  0.8549614025111897
Accuracy Percentage : 93.37%
```

In [112…

```
results = pd.DataFrame({
    'Model': ['Linear Regression', 'Random Forest', 'Gradient Boosting', 'XGBoost', 'CatBoost', 'AdaBoost', 'Lig
    'Mean Squared Error (MSE)': [MSE_lr,MSE_rf, MSE_gb, MSE_xgb, MSE_cb, MSE_ab, MSE_lgb, MSE_ngb, MSE_sr ],
    'R-squared score (R2)': [R2_Score_lr,R2_Score_rf,R2_Score_gb, R2_Score_xgb, R2_Score_cb, R2_Score_ab, R2_Sc
    'Accuracy ': [Accuracy_lr, Accuracy_rf, Accuracy_gb, Accuracy_xgb, Accuracy_cb, Accuracy_ab, Accuracy_lgb, A
})
print(results)
```

```
                Model  Mean Squared Error (MSE)  R-squared score (R2)  \
0   Linear Regression                   0.412833              0.799987
1       Random Forest                   0.319772              0.845074
2   Gradient Boosting                   0.313416              0.848153
3             XGBoost                   0.324266              0.842896
4            CatBoost                   0.269770              0.869299
5            AdaBoost                   0.510858              0.752495
6            LightGBM                   0.290256              0.859374
7             NGBoost                   0.330229              0.840008
8  Stacking Regressor                   0.299364              0.854961

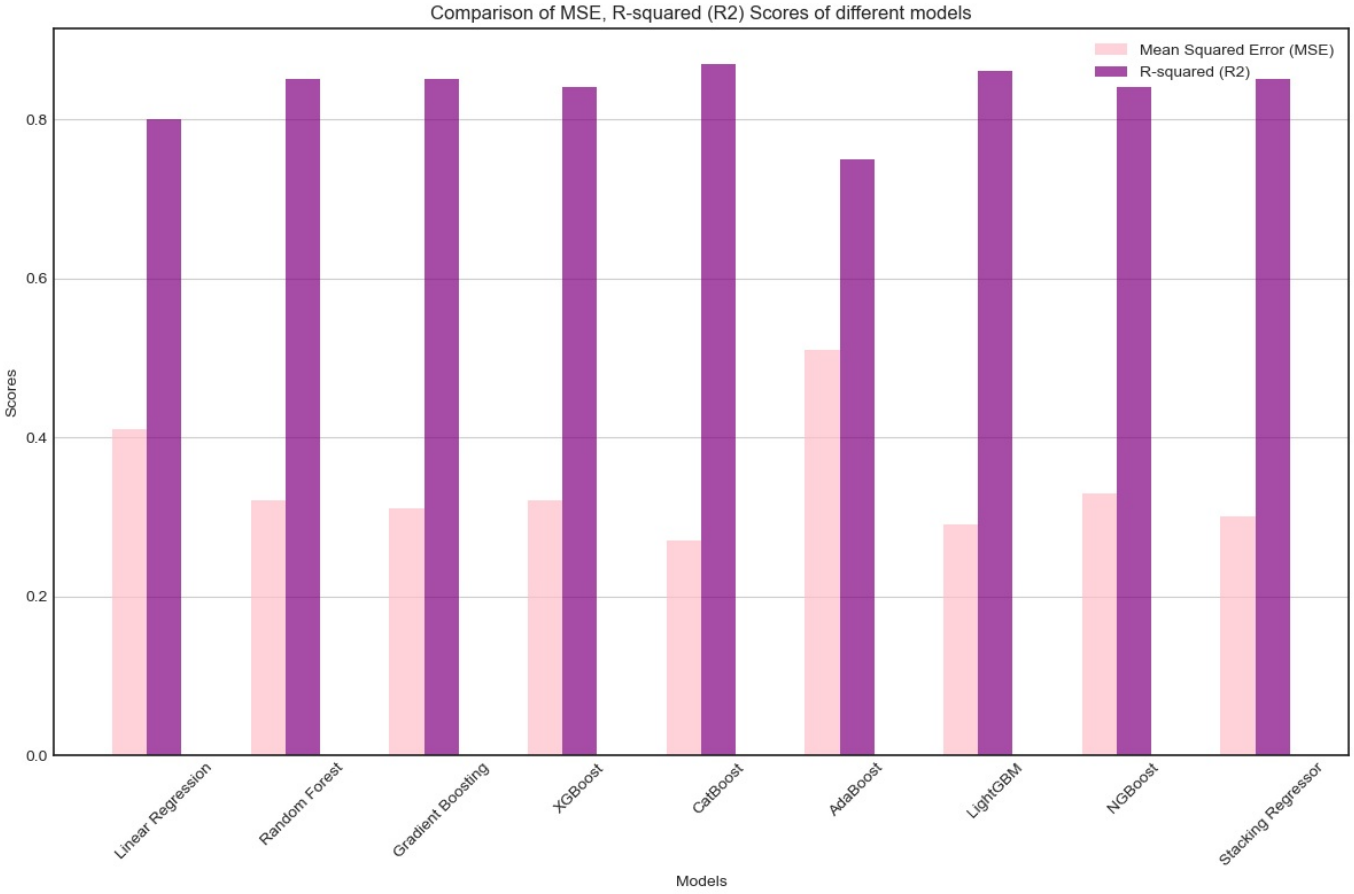
   Accuracy
0  91.932209
1  93.453645
2  92.999244
3  93.201390
4  93.770054
5  90.835013
6  93.542846
7  92.866065
8  93.371194
```

In [114…
```
models = results['Model']
mse_scores = results['Mean Squared Error (MSE)'].round(2)
r2_scores = results['R-squared score (R2)'].round(2)
num_models = len(models)
bar_width = 0.25
index = np.arange(num_models)
```

In [116…
```
plt.figure(figsize=(12, 8))

plt.bar(index, mse_scores, bar_width, label='Mean Squared Error (MSE)', color='pink', alpha=0.7)
plt.bar(index + bar_width, r2_scores, bar_width, label='R-squared (R2)', color='purple', alpha=0.7)

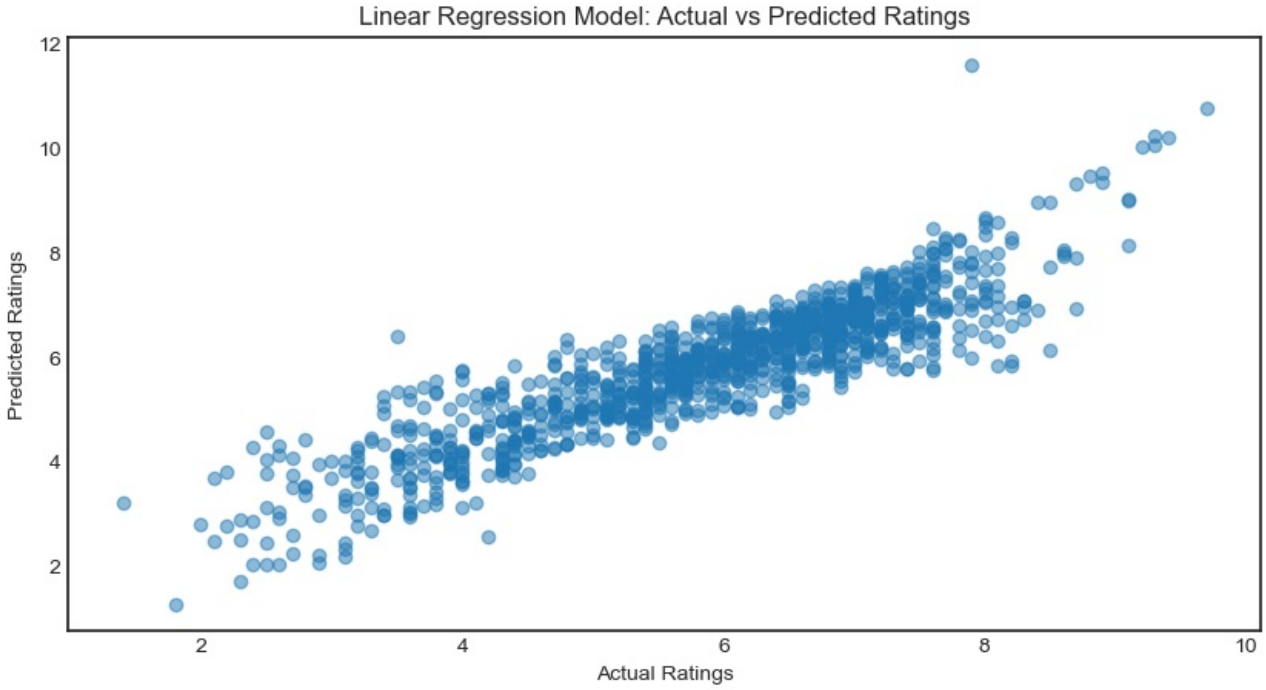
plt.xlabel('Models')
plt.ylabel('Scores')
plt.title('Comparison of MSE, R-squared (R2) Scores of different models')
plt.xticks(index + bar_width, models, rotation=45)
plt.legend()
plt.grid(axis='y')
plt.tight_layout()
plt.show()
```

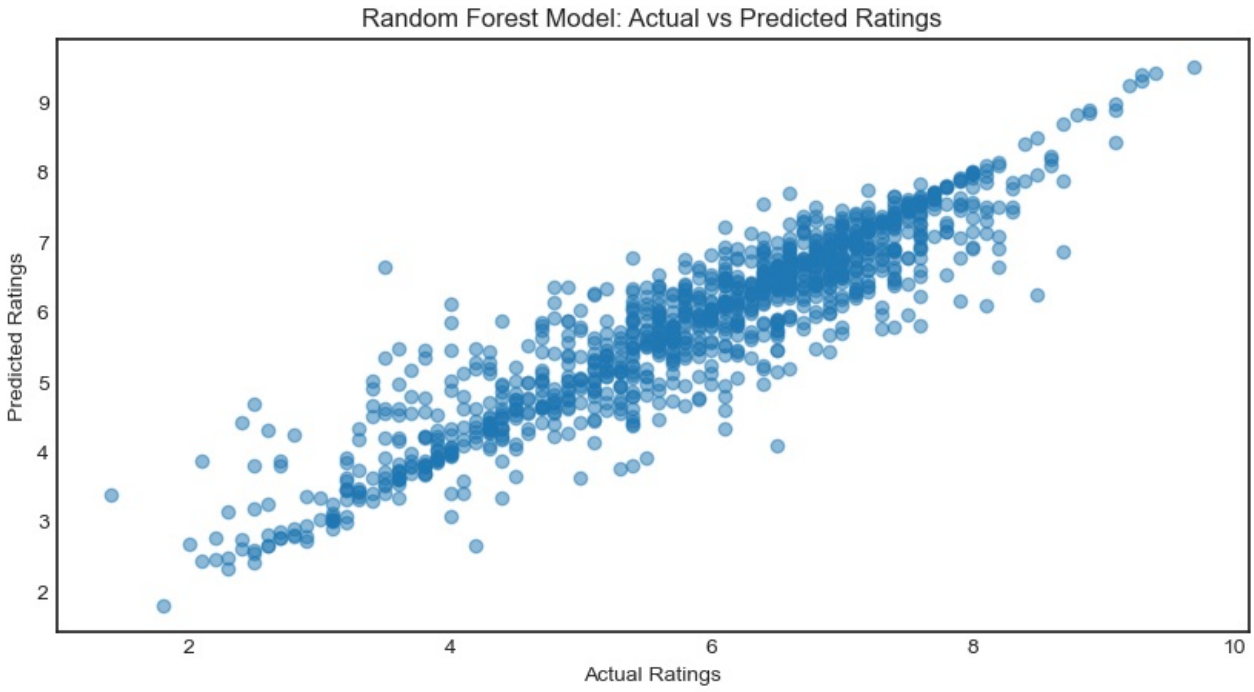


In [118…
```
plt.figure(figsize=(10, 5))
plt.scatter(y_test, y_pred_lr, alpha=0.5)
plt.title('Linear Regression Model: Actual vs Predicted Ratings')
plt.xlabel('Actual Ratings')
plt.ylabel('Predicted Ratings')
```

```
plt.show()

plt.figure(figsize=(10, 5))
plt.scatter(y_test, y_pred_rf, alpha=0.5)
plt.title('Random Forest Model: Actual vs Predicted Ratings')
plt.xlabel('Actual Ratings')
plt.ylabel('Predicted Ratings')
plt.show()
```





In [120…

```
x.head(10)
```

Out[120…

	Year	Votes	Duration	Genre_Average_Rating	Director_Average_Rating	Actor1_Average_Rating	Actor2_Average_Rating	Actor3
1	2019	8	109	6.415521	7.000000	6.850000	7.000000	
3	2019	35	110	5.716822	4.400000	5.420000	4.400000	
5	1997	827	147	6.242222	5.313333	4.788889	5.786667	
6	2005	1086	142	6.820000	7.383333	5.435000	6.933333	
8	2012	326	82	5.477778	5.600000	5.600000	5.883333	
9	2014	11	116	5.337778	4.000000	4.000000	4.550000	
10	2004	17	96	6.415521	6.200000	5.480000	5.450000	
11	2016	59	120	4.648649	6.950000	6.066667	5.900000	
12	1991	983	161	5.400000	4.907143	5.187179	5.976190	
13	2012	512	166	5.765957	5.700000	6.300000	5.700000	

In [122…

```
y.head(10)
```

Out[122…

```
1     7.0
3     4.4
5     4.7
6     7.4
8     5.6
9     4.0
10    6.2
11    5.9
12    6.5
13    5.7
Name: Rating, dtype: float64
```

In [124…

```
data = {'Year': [2018], 'Votes': [100] ,'Duration': [130], 'Genre_Average_Rating':[7.0],'Director_Average_Ratin
trail_data = pd.DataFrame(data)
```

In [126…

```
predict_rating = fitted_model_rf.predict(trail_data)
print('Predicted Rating for trial data: ',predict_rating[0])
```

```
Predicted Rating for trial data:  7.659000000000003
```

In [ ]: